数理分析方法

欧氏空间的极限与拓扑

Lecture 106

原课件逐页保留 · 零基础中文讲解 · 证明训练

材料与编写说明

课程原页：张顺明教授课件（用户提供）。参考：Gérard Debreu, Theory of Value (1959)，第一章 Mathematics。原页与本讲义的补充、纠错、例题分开呈现。本文为辅助自学材料，不代表教师官方讲义。

怎样阅读

先看本页的中文说明，再对照原页符号；遇到证明，先抄出假设和目标，再逐行说明“这一步用的是什么”。遮住证明重写一遍，最后用反例检查条件是否必要。重复或动画展开页仍逐页保留；说明会指出新增步骤或对应的完整论证位置。

“补充”用于补基础或展开原课件略去的论证；“纠错”指出原页需要修正的地方。课程概览及人物介绍页按其原有用途解释，不虚构数学定理。

编写日期：2026 年 10 月 24 日

Contents

零基础读证明：随手查阅 4
1 原课件第 1 页：本讲主题：有限维空间中的极限 6
2 原课件第 2 页：本讲在整门课中的位置 7
3 原课件第 3 页：一个点就是一串有顺序的实数 8
4 原课件第 4 页：二维坐标的几何读法 9
5 原课件第 5 页：三维坐标与二维切面 10
6 原课件第 6 页：坐标子空间与维数 11
7 原课件第 7 页：三种距离与一般的 p 距离 12
8 原课件第 8 页：为什么 $p \to \infty$ 会得到最大坐标 13
9 原课件第 9 页：距离公理与距离函数的连续性 14
10 原课件第 10 页：开球、闭球、去心邻域 15
11 原课件第 11 页：范数是到原点的距离 16
12 原课件第 12 页：范数比较与三角不等式的依据 17
13 原课件第 13 页：有界：一个统一的围栏 18
14 原课件第 14 页：向量收敛与逐坐标收敛 19
15 原课件第 15 页：极限唯一与收敛必有界 20
16 原课件第 16 页：内点、外点、边界点：量词不能写错 21
17 原课件第 17 页：聚点、导集、内部、闭包 22
18 原课件第 18 页：聚点的数列判别：构造互不相同的点 23
19 原课件第 19 页：导集的单调性与有限并公式 24
20 原课件第 20 页：内部的单调性与有限交公式 25
21 原课件第 21 页：Bolzano–Weierstrass：有界无限集必有聚点 26
22 原课件第 22 页：二分立方体证明：不断保留无限多个点 27
23 原课件第 23 页：二分证明的收尾：共同点就是聚点 28
24 原课件第 24 页：孤立点与可数性的证明起点 29
25 原课件第 25 页：把孤立点注入有理坐标点 30
26 原课件第 26 页：聚点也可以用去心邻域定义 31
27 原课件第 27 页：一个额外点为什么能推出无限多个点 32
28 原课件第 28 页：邻域换中心与内部的局部稳定性 33
29 原课件第 29 页：闭集的定义与三个基本例子 34
30 原课件第 30 页：连续性与上下水平集闭：补上子列步骤 35
31 原课件第 31 页：开球的闭包与稠密 36
32 原课件第 32 页：抽象闭包运算：四条规则 37

33 原课件第 33 页：导集与闭包本身都是闭集 38
34 原课件第 34 页：有限个闭集之并仍闭：导集方法 39
35 原课件第 35 页：同一个有限并结论：闭包方法 40
36 原课件第 36 页：可数个闭集之交仍闭：导集方法 41
37 原课件第 37 页：可数闭交的闭包写法 42
38 原课件第 38 页：任意闭集族的交：不需要数得出来 43
39 原课件第 39 页：任意闭交的闭包证明与笔误 44
40 原课件第 40 页：为什么无限并会出现新极限点 45
41 原课件第 41 页：闭包是包含原集合的最小闭集 46
42 原课件第 42 页：递减非空紧集的交非空：简洁完整证明 47
43 原课件第 43 页：闭集习题详解 48
44 原课件第 44 页：内点与内部 49
45 原课件第 45 页：内点与内部 50
46 原课件第 46 页：内点与内部 51
47 原课件第 47 页：有限开集的交 52
48 原课件第 48 页：有限开集的交 53
49 原课件第 49 页：任意开集的并 54
50 原课件第 50 页：任意开集的并 55
51 原课件第 51 页：任意开集的并 56
52 原课件第 52 页：任意开集的并 57
53 原课件第 53 页：实开集的区间分解 58
54 原课件第 54 页：实开集的区间分解 59
55 原课件第 55 页：欧氏开集的方块分解 60
56 原课件第 56 页：欧氏开集的方块分解 61
57 原课件第 57 页：可数拓扑基 62
58 原课件第 58 页：开集习题与必要纠错 63
59 原课件第 59 页：紧性与有限子覆盖 64
60 原课件第 60 页：紧性与有限子覆盖 65
61 原课件第 61 页：紧性与有限子覆盖 66
62 原课件第 62 页：紧性与有限子覆盖 67
63 原课件第 63 页：紧性与有限子覆盖 68
64 原课件第 64 页：紧性与有限子覆盖 69
65 原课件第 65 页：紧集的邻域与逆定理 70
66 原课件第 66 页：紧集的邻域与逆定理 71

67 原课件第 67 页：紧性习题 72
68 原课件第 68 页：闭包与内部的抽象公理 73
69 原课件第 69 页：闭包与内部的抽象公理 74
70 原课件第 70 页：闭包与内部的抽象公理 75
71 原课件第 71 页：闭包与内部的抽象公理 76
72 原课件第 72 页：闭包与内部的抽象公理 77
73 原课件第 73 页：闭包与内部的抽象公理 78
74 原课件第 74 页：闭包与内部的抽象公理 79
75 原课件第 75 页：闭包与内部的抽象公理 80
76 原课件第 76 页：闭包的序列判据与相对闭性 81
77 原课件第 77 页：闭包的序列判据与相对闭性 82
78 原课件第 78 页：闭包的序列判据与相对闭性 83
79 原课件第 79 页：闭包的序列判据与相对闭性 84
80 原课件第 80 页：闭包的序列判据与相对闭性 85
81 原课件第 81 页：闭包的序列判据与相对闭性 86
82 原课件第 82 页：闭包的序列判据与相对闭性 87
83 原课件第 83 页：闭包的序列判据与相对闭性 88
84 原课件第 84 页：任意欧氏子集都可分 89
85 原课件第 85 页：任意欧氏子集都可分 90
86 原课件第 86 页：任意欧氏子集都可分 91
87 原课件第 87 页：内部、边界与外部 92
88 原课件第 88 页：内部、边界与外部 93
89 原课件第 89 页：内部、边界与外部 94
90 原课件第 90 页：内部、边界与外部 95
91 原课件第 91 页：有界、紧与有限积 96
92 原课件第 92 页：有界、紧与有限积 97
93 原课件第 93 页：连通的多种判据 98
94 原课件第 94 页：连通的多种判据 99
95 原课件第 95 页：连通的多种判据 100
96 原课件第 96 页：连通的多种判据 101
97 原课件第 97 页：连通的多种判据 102
98 原课件第 98 页：连通的多种判据 103
99 原课件第 99 页：连通的多种判据 104
100 原课件第 100 页：连通的多种判据 105

零基础读证明：随手查阅

对象的层次。 $x \in A$ 表示一个元素属于集合，$A \subseteq B$ 表示 A 的每个元素都在 B 中。$\{x\}$ 是只含 x 的集合，不是 x 本身。$\varnothing$ 是空集，$\{\varnothing\}$ 则含有一个元素。$\mathbb{R}^N$ 中的点是有 N 个坐标的有序列表。

量词。 $\forall$ 是“对所有”，$\exists$ 是“至少存在一个”。例如“每个学生有一本书”不要求是同一本；“有一本书供每个学生使用”则要求同一本。证明极限时 $\forall \varepsilon > 0\ \exists n_0\ \forall n \geq n_0$ 的顺序不可换：先指定精度，再找统一起点，最后控制这个起点之后的全部项。

怎样否定。 否定“所有 x 满足 P”得到“存在 x 不满足 P”；否定“存在”得到“所有都不”。否定 $P \Rightarrow Q$ 是 P 真且 Q 假。因此要推翻一个定理，必须满足其全部假设又违反结论；拿不满足假设的例子不能当反例。

四种基本证明。 (1) 包含 $A \subseteq B$：任取 $x \in A$，用定义得到 $x \in B$。(2) 相等 $A = B$：两个方向分别证明。(3) 等价 $P \iff Q$：证明 $P \Rightarrow Q$ 及反向。(4) 反证：把结论否定加入已知，推出矛盾。逆否命题 $\neg Q \Rightarrow \neg P$ 与 $P \Rightarrow Q$ 等价，“反过来” $Q \Rightarrow P$ 则未必成立。

数列、子列和收敛。 数列 x_n 是第 n 个对象；子列 x_{n_k} 的指标必须严格递增。$x_n \to x$ 指任意小的误差最终都能保持。前面有限多项不影响极限。抽子列通常是为了用有界性和有限维 Bolzano–Weierstrass 定理获得极限；抽取之后须把原来所有相关量一起限制到同一子列。

距离的关键估计。 $\|x\| = (\sum_i x_i^2)^{1/2}$，$\|x - y\|$ 是距离。三角不等式 $\|x - z\| \leq \|x - y\| + \|y - z\|$ 允许经中间点拆误差。两段误差各小于 $\varepsilon/2$，总误差便小于 ε；分成 m 段时可各分配 ε/m。这里的 ε 不是固定的小数，而是任意正精度。

做证明题的五行草稿。 先写完整假设；写准确目标；列出可用定义与定理；逐步注明所用条件；最后检查任意性、存在性和量词顺序。出现“显然”时问：若阅卷者要求展开，我能否补出一个集合等价链、一个不等式或一个明确构造？不知道证明的命题要标为引用，不能把待证结论循环当作理由。

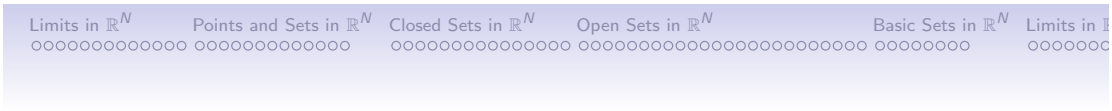
开始之前：把空间、极限、集合分开理解

本讲的核心问题是：一列点越来越接近某处时，那个地方是否仍在原集合里？这句话会依次变成极限、闭集、紧集，并为下一讲的连续性服务。全文在有限维欧氏空间讨论；维数记为 N，数列编号记为 k，不要把二者混淆。x^k 表示第 k 个向量，x_j^k 表示它的第 j 个坐标，通常不是乘方。

约定 $B_r(x)=\{y:\|y-x\|<r\}$ 为开球，$\overline{B}_r(x)=\{y:\|y-x\|\le r\}$ 为闭球。原课件对 N,B 的使用前后不完全一致，讲解统一采用这个约定。E' 是聚点集，$\overline{E}$ 是闭包，E° 是内部，∂E 是边界，E^c 是相对于当前空间的补集。撇号在此不是求导。

证明的三种基本动作　证明 $A\subseteq B$：任取 $x\in A$，推出 $x\in B$。证明等号：做两次包含。证明“任意 $\varepsilon>0$”的结论：先让别人任意给出一个正数 ε，然后依题意选择半径或编号；选择不能依赖尚未给出的变量。

1 原课件第 1 页：本讲主题：有限维空间中的极限



Theory of Value

An Axiomatic Analysis of Economic Equilibrium

Gerard Debreu (1959)

Professor Shunming Zhang

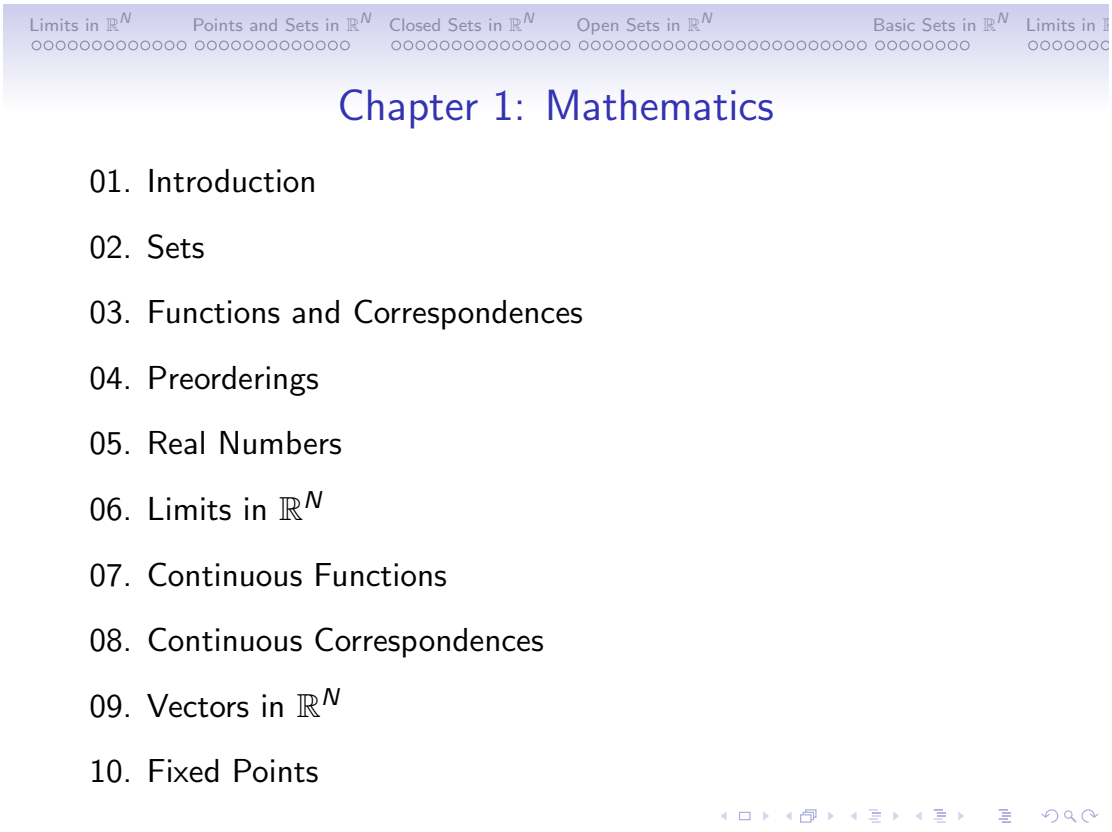
School of Finance
Renmin University of China

September 26, 2022

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页是课件封面。本讲围绕 $\mathbb{R}^N$ 中的点和点集展开，与 Debreu《Theory of Value》第一章第6节对应。学习目标不是记住所有图形，而是能用“邻域、数列、补集”三种语言表达同一件事。证明考试中，必须写清楚所取的点、半径、数列分别属于哪里。

2 原课件第 2 页：本讲在整门课中的位置



上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

第2节集合与第3节函数提供语言，第5节实数提供上确界和完备性，本节把实数结论推广到有限个坐标。后面连续函数需要闭集和紧集，连续对应需要序列与闭图，不动点定理需要非空、紧、凸。因此本节优先掌握：极限的量词、闭集的序列判别、紧性的有限覆盖性质，以及连通与区间的关系。

3 原课件第 3 页：一个点就是一串有顺序的实数



The set $\mathbb{R}^N$ is the product of N sets equal to $\mathbb{R}$, i.e, $\mathbb{R} \times \cdots \times \mathbb{R} \times \cdots \times \mathbb{R}$. According to the general definitions of 1.2.11, an element, or point, x of $\mathbb{R}^N$ is an N-tuple of real numbers $x = (x_n) = (x_1, \cdots, x_n, \cdots, x_N)$. The nth number in the N-tuple is the nth coordinate of x.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$\mathbb{R}^N = \underbrace{\mathbb{R} \times \cdots \times \mathbb{R}}_{N\text{ 个}}$。例如 $(2, -1, 4) \in \mathbb{R}^3$，其第2坐标是 -1。坐标有顺序，$(1, 2)$ 与 $(2, 1)$ 不同。经济学里一个商品组合可以记成 $x = (x_1, \ldots, x_N)$，但本讲暂不要求坐标非负。证明中写 $x = y$ 等价于逐一证明 $x_j = y_j$ 对所有 j 成立。

4 原课件第 4 页：二维坐标的几何读法



The set $\mathbb{R}^2$ may be visualized as follows. Draw in a plane a horizontal and a vertical straight line. The first (resp. second) set $\mathbb{R}$ in $\mathbb{R} \times \mathbb{R}$ will be represented by the horizontal (resp. vertical) line by choosing 0 at the intersection point, and 1 arbitrarily to the right of 0 (resp. above 0). Then an element (x_1, x_2) of $\mathbb{R}^2$ is represented by the intersection point of the vertical through x_1 and the horizontal through x_2.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$\mathbb{R}^2$ 中第一坐标控制左右，第二坐标控制上下。点 (a,b) 是直线 $x_1 = a$ 与 $x_2 = b$ 的交点。区分点 (a,b) 与区间 $(a,b) \subset \mathbb{R}$：同一括号的意思由上下文决定。以后讨论一个平面集合是否开，必须允许向所有平面方向作小移动，不能只沿一条线移动。

5 原课件第 5 页：三维坐标与二维切面



To visualize $\mathbb{R}^3$, consider, in ordinary space, the west-east, the south-north, and the vertical straight line through a point 0. They will represent respectively the first, the second, and the third set in $\mathbb{R} \times \mathbb{R} \times \mathbb{R}$. An element (x_1, x_2, x_3) of $\mathbb{R}^3$ is represented by the intersection point of the vertical south-north plane through x_1, the vertical west-east plane through x_2, and the horizontal plane through x_3.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$\mathbb{R}^3$ 中一个等式 $x_1 = a$ 通常确定一个平面，三个独立坐标等式才确定一个点。这样理解比想象高维图形更可靠：高维证明基本上都是逐坐标计算。例如 $x = (1, 0, -2)$，其第三坐标是 -2；保持其他坐标不动、只改变第三坐标，仍在 $\mathbb{R}^3$ 中。

6 原课件第 6 页：坐标子空间与维数

Limits in $\mathbb{R}^N$

$\mathbb{R}^N$ is called the *N*-dimensional Euclidean space. Let $\mathscr{N}$ be a subset of the set of the first N positive integers $\{1, \cdots, n, \cdots, N\}$. The set $\left\{x \in \mathbb{R}^N \mid x_n = 0, n \notin \mathscr{N}\right\}$ is called a coordinate subspace of $\mathbb{R}^N$. Its dimension is the number of elements of $\mathscr{N}$. The point $\mathbf{0}$, all of whose coordinates are equal to 0, is the origin of $\mathbb{R}^N$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

给定指标集合 $I \subseteq \{1, \ldots, N\}$，只允许 $j \in I$ 的坐标自由变化，其余坐标固定为0，得到坐标子空间，维数为 $|I|$。例如在 $\mathbb{R}^3$ 中令 $I = \{1, 3\}$，得到 $\{(a, 0, c) : a, c \in \mathbb{R}\}$，是二维平面。原点 $0 = (0, \ldots, 0)$ 不是空集；一个是点，一个是没有元素的集合。

7 原课件第 7 页：三种距离与一般的 p 距离

Sets in $\mathbb{R}^N$

N-dimensional Euclidean space, or N-dimensional continuum, is the set $\mathbb{R}^N$ of all number tuple of N real numbers.

$$\mathbb{R}^N = \{x = (x_1, \cdots, x_N) : x_n \in \mathbb{R} \text{ for } n = 1, \cdots N\}.$$

In general, for any two points $x^k = (x_1^k, \cdots, x_N^k)$ in $\mathbb{R}^N$ for $k = 1, 2$, we define the distance (Euclidean l^p metric) between the two points for $1 \leqslant p < \infty$

$$\rho_p(x^1, x^2) = \left\{ \sum_{n=1}^{N} \left| x_n^1 - x_n^2 \right|^p \right\}^{\frac{1}{p}}.$$

The useful metrics on $\mathbb{R}^N$ include the taxicab (or l^1) metric for $p = 1$

$$\rho_1(x^1, x^2) = \sum_{n=1}^{N} \left| x_n^1 - x_n^2 \right|,$$

the distance (Euclidean l^2 metric) between the two points for $p = 2$

$$\rho_2(x^1, x^2) = \left\{ \sum_{n=1}^{N} \left(x_n^1 - x_n^2 \right)^2 \right\}^{\frac{1}{2}},$$

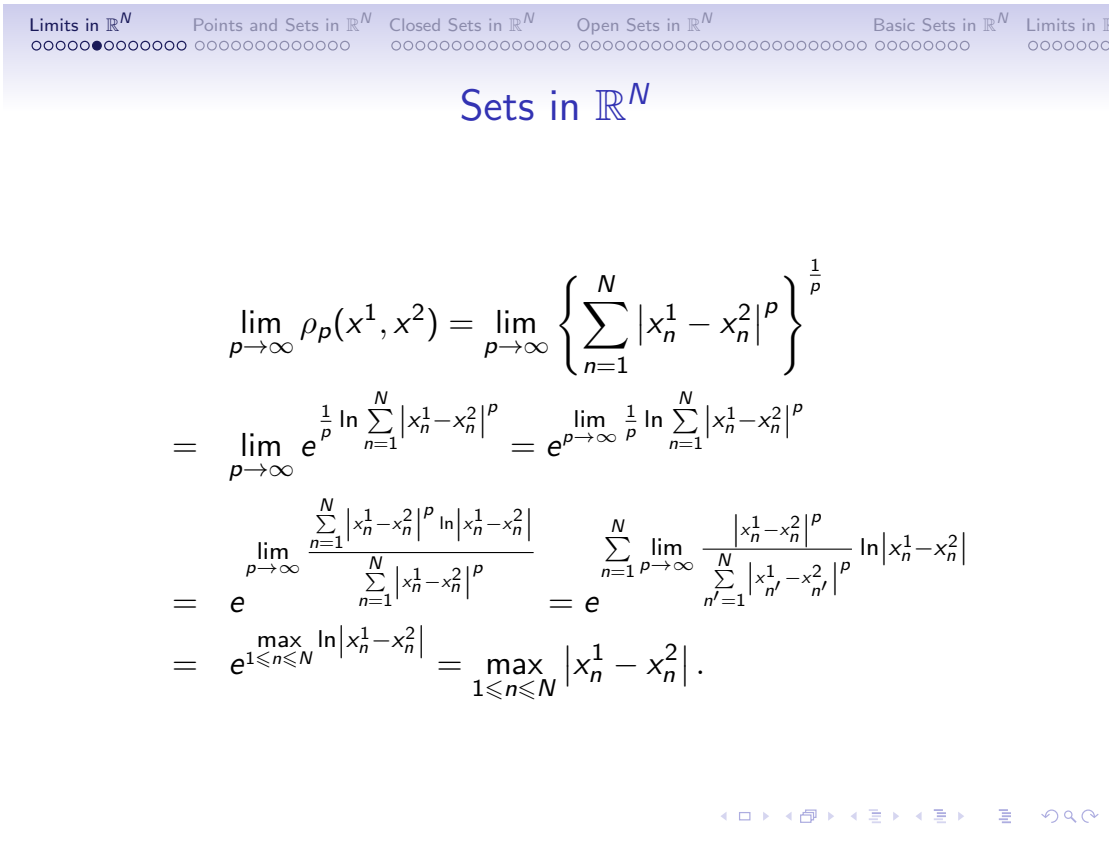
and the max (or l^∞) metric for $p = \infty$

$$\rho_\infty(x^1, x^2) = \max \left\{ \left| x_n^1 - x_n^2 \right| : 1 \leqslant n \leqslant N \right\}.$$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

令 $a_j = |x_j - y_j|$，则 $\rho_1(x,y) = \sum a_j$，$\rho_2(x,y) = \sqrt{\sum a_j^2}$，$\rho_\infty(x,y) = \max_j a_j$。从 $(0,0)$ 到 $(3,4)$，三者分别为7、5、4。p 距离是 $(\sum a_j^p)^{1/p}$；$p \geq 1$ 才有课件要求的三角不等式。所谓 ℓ^p 是距离的名称，通常只有 $p = 2$ 被称为欧氏距离。后面的开闭性在这些距离下相同，但距离数值并不相同。

8 原课件第 8 页：为什么 $p \to \infty$ 会得到最大坐标



上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

课件用对数和洛必达法则计算；含有 $a_j = 0$ 时 $\log a_j$ 需要额外处理。以下夹逼证明更短且覆盖所有情形。令 $M = \max_j a_j$。因为某个 $a_j = M$，而每个 $a_j \le M$，有

$$M^p \le \sum_{j=1}^{N} a_j^p \le N M^p, \qquad M \le \rho_p(x,y) \le N^{1/p} M.$$

$N^{1/p} \to 1$，故夹逼得 $\rho_p \to M = \rho_\infty$。若 $M = 0$，左右本来全为0。证明模板是“最大项给下界，项数乘最大项给上界”。

9 原课件第 9 页：距离公理与距离函数的连续性



Then we have the following three propositions.

(1) Positivity: $\rho(x^1, x^2) \geqslant 0$. $\rho(x^1, x^2) = 0$ if and only if $x^1 = x^2$ (i.e. $x_n^1 = x_n^2$ for $n = 1, \cdots N$).

(2) Symmetry: $\rho(x^1, x^2) = \rho(x^2, x^1)$.

(3) Triangle Inequality: $\rho(x^1, x^2) \leqslant \rho(x^1, x^0) + \rho(x^0, x^2)$.

From the triangle inequality, $\rho(x^1, x^2)$ is a continuous function on (x^1, x^2): If $\lim_{n\to\infty} \rho(x_n^1, x^1) = 0$ and $\lim_{n\to\infty} \rho(x_n^2, x^2) = 0$, then $\lim_{n\to\infty} \rho(x_n^1, x_n^2) = \rho(x^1, x^2)$. In particular, $\lim_{\rho(x_n^2, x^2)\to 0} \rho(x^1, x_n^2) = \rho(x^1, x^2)$ and $\lim_{\rho(x_n^1, x^1)\to 0} \rho(x_n^1, x^2) = \rho(x^1, x^2)$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

非负性与对称性直接来自绝对值；三角不等式表示绕经第三点不会更短。由两次三角不等式得

$$|\rho(x,y) - \rho(u,v)| \le \rho(x,u) + \rho(y,v).$$

具体地，$\rho(x,y) \le \rho(x,u)+\rho(u,v)+\rho(v,y)$ 给一个方向，再交换两对点给反方向。如果 $x_k \to x, y_k \to y$，右边趋0，于是 $\rho(x_k,y_k) \to \rho(x,y)$。这证明不依赖尚未学过的连续函数运算法则，是后面使用距离连续性的依据。一般 p 距离的三角不等式推导见第12页讲解。

10 原课件第 10 页：开球、闭球、去心邻域

Sets in $\mathbb{R}^N$

The set of all points in $\mathbb{R}^N$ whose distance to some given point x_0 is strictly less than δ is call neighborhood with center x_0 and radius δ (without boundary), written as $N(x_0, \delta)$.

$$N(x_0, \delta) = \{x \in \mathbb{R}^N : \rho(x, x_0) < \delta\}.$$

When it must not mention point x_0 or radius δ, we just say a neighborhood. If the boundary is required, we define the neighborhood with boundary. The set of all points in $\mathbb{R}^N$ whose distance to some given point x_0 is less than δ is call neighborhood with center x_0 and radius δ (with boundary), written as $B(x_0, \delta)$.

$$B(x_0, \delta) = \{x \in \mathbb{R}^N : \rho(x, x_0) \leqslant \delta\}.$$

Sometimes we do not need the center x_0 in neighborhood with center x_0 and radius δ, $B(x_0, \delta)$. The set of all points in $\mathbb{R}^N$ whose distance to some given point x_0 is strictly less than δ is call neighborhood with center x_0 and radius δ without center x_0, written as $D(x_0, \delta)$.

$$\begin{aligned} D(x_0, \delta) &= \{x \in \mathbb{R}^N : 0 < \rho(x, x_0) < \delta\} \\ &= \{x \in \mathbb{R}^N : \rho(x, x_0) < \delta, x \neq x_0\} \\ &= N(x_0, \delta) \setminus \{x_0\} \end{aligned}$$

which is neighborhood with center x_0 and radius δ deleting center x_0.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

开球要求距离严格小于 r；闭球允许等于 r；去心球 $B_r(x) \setminus \{x\}$ 额外排除中心。半径始终要求 $r > 0$。例如在实数线上，开球是 $(x - r, x + r)$，闭球是 $[x - r, x + r]$。在最大范数下球是方块，在欧氏范数下是圆或球。去心条件用于聚点：集合只有中心一个点时，不能因此称中心为聚点。

11 原课件第 11 页：范数是到原点的距离

Sets in $\mathbb{R}^N$

For any point $x = (x_1, \cdots, x_N)$ in $\mathbb{R}^N$, define its norm or metric $\|x\|$ to present $\rho(x, 0)$, the distance (Euclidean l^p metric) for $1 \leqslant p < \infty$

$$\|x\|_p = \rho_p(x, 0) = \left\{ \sum_{n=1}^{N} |x_n|^p \right\}^{\frac{1}{p}}.$$

The useful metrics on $\mathbb{R}^N$ include the taxicab (or l^1) metric for $p = 1$

$$\|x\|_1 = \rho_1(x, 0) = \sum_{n=1}^{N} |x_n|,$$

Euclidean l^2 metric for $p = 2$

$$\|x\|_2 = \rho_2(x, 0) = \left\{ \sum_{n=1}^{N} [x_n]^2 \right\}^{\frac{1}{2}},$$

and the max (or l^∞) metric for $p = \infty$

$$\|x\|_\infty = \rho_\infty(x, 0) = \max\{|x_n| : 1 \leqslant n \leqslant N\},$$

i.e., the greatest of the absolute values of its coordinates

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$\|x\|_p = \rho_p(x,0)$ 是向量的大小，$\rho_p(x,y) = \|x-y\|_p$ 是两个点的距离。$\|x\|_\infty = \max_j |x_j|$，不能漏掉绝对值。例如 $(-5,2)$ 的最大范数是5，不能取2。范数的输入是一个向量，距离的输入是两个点；这一区分能避免写错极限条件。

12 原课件第 12 页：范数比较与三角不等式的依据

Sets in $\mathbb{R}^N$

$$\|x\|_{p'} \leqslant \|x\|_p \leqslant N^{\frac{1}{p}-\frac{1}{p'}}\|x\|_{p'}, \qquad 1 \leqslant p \leqslant p' \leqslant \infty$$

$$\|x\|_{p'} \leqslant \|x\|_1 \leqslant N^{1-\frac{1}{p'}}\|x\|_{p'}, \qquad 1 = p < p' \leqslant \infty$$
$$\|x\|_{p'} \leqslant \|x\|_p \leqslant N^{\frac{1}{p}-\frac{1}{p'}}\|x\|_{p'}, \qquad 1 < p < p' < \infty$$
$$\|x\|_{\infty} \leqslant \|x\|_p \leqslant N^{\frac{1}{p}}\|x\|_{\infty}, \qquad 1 \leqslant p < p' = \infty$$

Then we have the following three propositions.

(1) Positivity: $\|x\| \geqslant 0$. $\|x\| = 0$ if and only if $x = 0$ (i.e. $x_n = 0$ for $n = 1, \cdots N$).

(2) Homogeneity (of Degree 1): $\|\alpha x\| = |\alpha|\|x\|$.

(3) Triangle Inequality: $\|x^1 + x^2\| \leqslant \|x^1\| + \|x^2\|$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

比较不等式的证明 设 $1 \le p \le q < \infty$。若 $x \neq 0$，令 $b_j = |x_j| / \|x\|_p$，则 $0 \le b_j \le 1$ 且 $\sum b_j^p = 1$。于是 $\sum b_j^q \le 1$，得到 $\|x\|_q \le \|x\|_p$。另一方面，对凹函数 $t^{p/q}$ 的平均值不等式给出

$$\frac{1}{N}\sum |x_j|^p \le \left(\frac{1}{N}\sum |x_j|^q\right)^{p/q}, \quad \|x\|_p \le N^{1/p-1/q}\,\|x\|_q .$$

$q = \infty$ 时直接用第8页的最大项估计。$x = 0$ 时显然成立。

三条范数公理 非负且仅零向量范数为0；$\|\alpha x\| = |\alpha|\,\|x\|$；$\|x+y\| \le \|x\| + \|y\|$。对 $p = 1, \infty$，逐坐标使用 $|x_j + y_j| \le |x_j| + |y_j|$ 即可。对 $p = 2$，先由 $\sum_j (x_j - t y_j)^2 \ge 0$ 的判别式推出柯西不等式 $|\sum x_j y_j| \le \|x\|_2\,\|y\|_2$，再展开 $\|x+y\|_2^2$ 即得结论。

一般 $p > 1$ 的补充证明 令 $q = p/(p-1)$。Young 不等式 $ab \le a^p/p + b^q/q$ 可由函数 $a^p/p - ab + b^q/q$ 对 $a \ge 0$ 的最小值为0证明。将两个向量分别除以其 p、q 范数，逐项应用并求和，就得 Hölder 不等式 $\sum |u_j v_j| \le \|u\|_p\,\|v\|_q$。记 $S = \sum |x_j + y_j|^p$，则

$$S \le \sum (|x_j| + |y_j|)|x_j + y_j|^{p-1} \le (\|x\|_p + \|y\|_p) S^{1/q}.$$

$S > 0$ 时除以 $S^{1/q}$ 得三角不等式；$S = 0$ 时显然成立。这些补充只用于解释公理的来源，后续可直接使用。

13 原课件第 13 页：有界：一个统一的围栏



E is a point set of $\mathbb{R}^N$, if there exists a constant M such that, for any $x = (x_1, \cdots, x_N) \in E$, $\|x\| \leqslant M$, then we say that E is a bounded set. For example, for $1 \leqslant p \leqslant \infty$, $\|x\|_p = \left\{\sum_{n=1}^N |x_n|^p\right\}^{\frac{1}{p}} \leqslant M$ for any $x = (x_1, \cdots, x_N) \in E$, then E is a bounded set. If $p = \infty$, E is a bounded set if and only if, for $x \in E$, there exists a constant M such that $\|x\|_\infty = \max\{|x_1|, \cdots, |x_N|\} \leqslant M$, that is, $|x_n| \leqslant M$ for $n = 1, \cdots, N$. Let M be a non-negative real number; the set $\mathbb{M} = \left\{x \in \mathbb{R}^N \mid \|x\| \leqslant M\right\}$ is a closed cube of $\mathbb{R}^N$ with center 0, edge $2M$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

集合 E 有界是 $\exists M < \infty\ \forall x \in E, \|x\| \leq M$。必须先选一个 M 管住整个集合；“对每个点存在它自己的 M”对任何集合都成立，不能表达有界。最大范数下 $\|x\|_\infty \leq M$ 等价于每个坐标在 $[-M, M]$ 中，因此得到边长 $2M$ 的闭立方体。**注意**只有最大范数的球是该立方体；欧氏范数的球不是立方体。有限维范数比较说明选哪一种范数不影响有界性。

14 原课件第 14 页：向量收敛与逐坐标收敛

Limits in $\mathbb{R}^N$

Consider a sequence (x^q) of points of $\mathbb{R}^N$. One says that (x^q) converges (or tends) to a point x^0 of $\mathbb{R}^N$ if, for any $\varepsilon > 0$, there exists a large integer number $Q > 0$ such that, when $q > Q$, $\|x^q - x^0\| < \varepsilon$. One writes $x^q \to x^0$.

For the norm or metric $\|x\|_p = \left\{\sum_{n=1}^N |x_n|^p\right\}^{\frac{1}{p}}$ for $1 \leqslant p \leqslant \infty$, the sequence (x^q) converges (or tends) to a point x^0 of $\mathbb{R}^N$ if, for any $\varepsilon > 0$, there exists a large integer number $Q > 0$ such that, when $q > Q$, $\|x^q - x^0\|_p < \varepsilon$ or equivalently $\|x^q - x^0\|_\infty = \max\left\{\left|x_1^q - x_1^0\right|, \cdots, \left|x_N^q - x_N^0\right|\right\} < \varepsilon$, that is, $\|x_n^q - x_n^0\|_\infty < \varepsilon$ for $n = 1, \cdots, N$, i.e., for all coordinates $(n = 1, \cdots, N)$, one has $x_n^q \to x_n^0$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$x^k \to x$ 的完整读法是：对任意误差 $\varepsilon > 0$，存在整数 K，使所有 $k \geq K$ 都满足 $\left\|x^k - x\right\| < \varepsilon$。允许 K 依赖误差，但不能只让某些很晚的项靠近。

等价性证明 向量收敛推出逐坐标收敛，因为 $|x_j^k - x_j| \leq \left\|x^k - x\right\|_\infty$。反过来，对每个 j 取 K_j 使误差小于 ε，再令 $K = \max_{1\leq j\leq N} K_j$，则全部坐标同时小于 ε。一般 p 范数时可先把坐标误差控制为 $\varepsilon/N^{1/p}$。课件中不同范数下同一个 ε 的不等式并非逐项等价；等价的是收敛性质。

15 原课件第 15 页：极限唯一与收敛必有界



As in the case of numbers (1.5.o), a sequence which tends to a point is called convergent; the unique point to which a convergent sequence tends is called its limit.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

若 $x^k \to a$ 且 $x^k \to b$，则 $\|a-b\| \le \|a-x^k\| + \|x^k-b\| \to 0$，所以 $a=b$。若想完全用量词：假设 $d=\|a-b\|>0$，取两边误差均小于 $d/3$ 就有 $d<2d/3$，矛盾。收敛数列一定有界：从某项后 $\|x^k-x\|<1$，于是尾部范数不超过 $\|x\|+1$；前面有限项的范数有最大值，把两者取最大即可。反过来有界不一定收敛，如 $(-1)^k$。

16 原课件第 16 页：内点、外点、边界点：量词不能写错

Accumulation Points, Interior Points, and Boundary Points

Let E be a set of N-dimensional Euclidean space $\mathbb{R}^N$ and P_0 be a given point in $\mathbb{R}^N$, we now explore the relationship between P_0 and E: there exist three possible settings.

(1) There are not any points of E around P_0. There exists a neighborhood $N(P_0, \delta)$ with center P_0 such that there are not any points of $N(P_0, \delta)$ in E: $N(P_0, \delta) \cap E = \varnothing$.

(2) All points around P_0 are in E. There exists a neighborhood $N(P_0, \delta)$ with center P_0 such that $N(P_0, \delta) \subseteq E$, then P_0 is one of interior points of E.

(3) Some points around P_0 are in E and some are not in. There exists a neighborhood $N(P_0, \delta)$ with center P_0 such that, for some $P \in N(P_0, \delta)$, $P \in E$, and for some $P' \in N(P_0, \delta)$, $P' \notin E$, then P_0 is one of boundary points of E.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

内点：存在一个球全部落在 E 中。外点：存在一个球完全不碰 E。边界点：每一个球既碰到 E 又碰到 E^c。**课件修正：**本页第(3)项写“存在一个混合邻域”不够；内点附近取很大的球也可能混合。例如 $E = (0, 1)$，点 $1/2$ 是内点，半径2的球却同时碰到 E 与补集。边界必须经得起任意缩小邻域。

17 原课件第 17 页：聚点、导集、内部、闭包

Accumulation Points, Interior Points, and Boundary Points

Let E be a set of $\mathbb{R}^N$ and P_0 be a given point in $\mathbb{R}^N$. If there exists a neighborhood $N(P_0, \delta)$ with center P_0 such that $N(P_0, \delta) \subseteq E$, then P_0 is one of interior points of E.

Let E be a set of $\mathbb{R}^N$ and P_0 be a given point in $\mathbb{R}^N$. If in any neighborhood $N(P_0, \delta)$ with center P_0 there exist infinite points in $N(P_0, \delta)$ belong to E, then P_0 is one of accumulation points (cluster points or limit points) of E.

Any interior point of E is an accumulation point of E. However, an accumulation point of E must not be a interior point of E. In fact, it is possible one of boundary points of E. Any interior point of E belongs to E. However, an accumulation point of E may belong to E, or may not belong to E.

Let E be a point set of $\mathbb{R}^N$. The set of all accumulation points of E is called the derived set, written as E'. The set of all interior points of E is called the interior, written as E°. $E \cup E'$ is call closure of E, written as $\overline{E} = E \cup E'$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

聚点 $x \in E'$：每个球中有无穷多个 E 的点。内点一定是聚点，因为非空开球含无穷多个点；聚点不必是内点，例如 $E = \{1/k : k \geq 1\}$ 的0。聚点也不必属于集合。内部 E° 收集内点；闭包 $\overline{E} = E \cup E'$ 将遗漏的极限位置补上。对上述 E，有 $E^\circ = \varnothing$，$E' = \{0\}$，$\overline{E} = E \cup \{0\}$。不要把闭包与导集混同：前者还保留每一个孤立点。

18 原课件第 18 页：聚点的数列判别：构造互不相同的点

Accumulation Points, Interior Points, and Boundary Points

Theorem 1. Let E be a set of $\mathbb{R}^N$ and P_0 be a given point in $\mathbb{R}^N$. $P_0 \in E'$ if and only if P_0 is one of limit points of E, or equivalently, there exists a sequence of different points $P_n \in E$ such that $\lim_{n\to\infty} \rho(P_n, P_0) = 0$.

Proof. [Sufficiency] For any neighborhood $N(P_0, \delta)$ of center P_0 with $\delta > 0$, then, if N is large enough, $\rho(P_n, P_0) < \delta$ for $n \geqslant N$, then $P_n \in N(P_0, \delta)$, and there exist infinite such P_n.

[Necessity] Define $\delta_n = \dfrac{1}{n}$, then $\delta_n > 0$. $P_0 \in E'$, and hence infinite points in $N(P_0, \delta_1)$ belong to E. We can take P_1 arbitrarily. $P_0 \in E'$, and hence infinite points in $N(P_0, \delta_2)$ belong to E, thus $N(P_0, \delta_2) \setminus \{P_1\} \neq 0$. We can take $P_2 \in N(P_0, \delta_2) \setminus \{P_1\}$ arbitrarily, both P_1 and P_2 are different. Generally, we assume that we take different $P_1, P_2, \cdots, P_N$ with $P_n \in N(P_0, \delta_n)$ and $P_n \in E$ for $n = 1, \cdots, N$. Then infinite points in $N(P_0, \delta_{N+1})$ belong to E, thus $N(P_0, \delta_{N+1}) \setminus \{P_1, P_2, \cdots, P_N\} \neq 0$. We can take $P_{N+1} \in N(P_0, \delta_{N+1}) \setminus \{P_1, P_2, \cdots, P_N\}$ arbitrarily, P_{N+1} is different from $P_1, P_2, \cdots, P_N$. Therefore, we obtain a sequence of different points $P_n \in E$ for $n = 1, 2, \cdots$ such that $P_n \in N(P_0, \delta_n)$ and $P_n \in E$ for $n = 1, 2, \cdots$. Note $\lim_{n\to\infty} \delta_n = 0$, then $\lim_{n\to\infty} \rho(P_n, P_0) = 0$.

Q.E.D.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

充分性 若互不相同的 $x_k \in E$ 趋于 x，任何球 $B_r(x)$ 都包含全部足够晚的项，因此包含无穷多个不同的点，故 $x \in E'$。

必要性 假设 $x \in E'$。从 $E \cap B_1(x)$ 选 $x_1 \neq x$；已选 $x_1, \ldots, x_k$ 后，从 $E \cap B_{1/(k+1)}(x)$ 中排除有限集合 $\{x, x_1, \ldots, x_k\}$，仍有点可选，记为 x_{k+1}。于是点互不相同，且 $\|x_k - x\| < 1/k \to 0$。

容易漏的条件 若允许数列恒为 x，只能说明 $x \in \overline{E}$，不能说明 $x \in E'$。“互不相同”是本定理区别于闭包判别的关键。

19 原课件第 19 页：导集的单调性与有限并公式

Accumulation Points, Interior Points, and Boundary Points

Theorem 2. If $A_1 \subseteq A_2$, then $A_1' \subseteq A_2'$ and $\overline{A}_1 \subseteq \overline{A}_2$.

Theorem 3. $(A_1 \cup A_2)' = A_1' \cup A_2'$.

Proof. Note $A_1 \subseteq A_1 \cup A_2$ and $A_2 \subseteq A_1 \cup A_2$, then $A_1' \subseteq (A_1 \cup A_2)'$ and $A_2' \subseteq (A_1 \cup A_2)'$ from Theorem 2, and hence $A_1' \cup A_2' \subseteq (A_1 \cup A_2)'$.

If $P \in (A_1 \cup A_2)'$, then there exists a sequence of different points $P_n \in A_1 \cup A_2$ such that $\lim\limits_{n\to\infty} \rho(P_n, P) = 0$. If $P \in A_1'$, then $P \in A_1' \cup A_2'$. If $P \notin A_1'$, then at most finite P_n for $n = 1, 2, \cdots$ belong to A_1, the remaining infinite points belong to A_2, then $P \in A_2'$ from Theorem 1. Thus $P \in A_1' \cup A_2'$. Therefore $(A_1 \cup A_2)' \subseteq A_1' \cup A_2'$. Q.E.D.

Corollary 1. $\left(\bigcup\limits_{n=1}^{N} A_n\right)' = \bigcup\limits_{n=1}^{N} A_n'$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$A \subseteq B$ 时，任意球中无穷多个 A 点也都是 B 点，所以 $A' \subseteq B'$，进而 $\overline{A} \subseteq \overline{B}$。要证 $(A \cup B)' = A' \cup B'$，容易的一边来自单调性。对另一边用逆否命题更清楚：若 $x \notin A'$ 且 $x \notin B'$，分别有半径 r, s 的球只含有限多个 A, B 点。半径 $\min(r, s)$ 的球只含有限多个 $A \cup B$ 点，故 $x \notin (A \cup B)'$。重复这个论证得到任意有限并的公式；无限并不能照搬，例见第43页第3题。

20 原课件第 20 页：内部的单调性与有限交公式

Accumulation Points, Interior Points, and Boundary Points

Theorem 4. If $A_1 \subseteq A_2$, then $A_1^\circ \subseteq A_2^\circ$.

Theorem 5. $(A_1 \cap A_2)^\circ = A_1^\circ \cap A_2^\circ$.

Proof. Note $A_1 \cap A_2 \subseteq A_1$ and $A_1 \cap A_2 \subseteq A_2$, then $(A_1 \cap A_2)^\circ \subseteq A_1^\circ$ and $(A_1 \cap A_2)^\circ \subseteq A_2^\circ$ from Theorem 2, and hence $(A_1 \cap A_2)^\circ \subseteq A_1^\circ \cap A_2^\circ$.

If $P \in A_1^\circ \cap A_2^\circ$, then $P \in A_k^\circ$ for $k = 1, 2$, and there exists a a neighborhood $N(P, \delta)$ of center P such that $N(P, \delta) \subseteq A_k$ for $k = 1, 2$, thus $N(P, \delta) \subseteq A_1 \cap A_2$. Therefore $P \in (A_1 \cap A_2)^\circ$, and hence $A_1^\circ \cap A_2^\circ \subseteq (A_1 \cap A_2)^\circ$. Q.E.D.

Corollary 2. $\left(\bigcap_{n=1}^{N} A_n \right)^\circ = \bigcap_{n=1}^{N} A_n^\circ$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

若 $A \subseteq B$，包含于 A 的小球也包含于 B，所以 $A^\circ \subseteq B^\circ$。由此 $(A \cap B)^\circ \subseteq A^\circ \cap B^\circ$。反向取 $x \in A^\circ \cap B^\circ$，存在 $r_A, r_B > 0$ 分别使小球落入两集合；令 $r = \min(r_A, r_B) > 0$，就有 $B_r(x) \subseteq A \cap B$。因此等号成立。 课件略去的步骤是把两个可能不同的半径合并成一个。有限交可以取有限个正数的最小值；无限多个正数的下确界可能为0，这正是无限交不保开性的原因。

21 原课件第 21 页：Bolzano–Weierstrass：有界无限集必有聚点

Bolzano-Weierstrass Theorem

Theorem 6 (Bolzano-Weierstrass Theorem). Let E be a bounded infinite set in N-dimensional Euclidean space $\mathbb{R}^N$. Then there exists at least an accumulation point P of E (The point P may not belong to E), i.e. $E' \neq \varnothing$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

定理同时需要有界与无限。$\mathbb{Z}$ 无限但无界，没有有限位置的聚点；$\{0,1\}$ 有界但有限，也没有聚点。聚点可能落在集合外，例如 $\{1/k\}$ 的聚点0。与数列版本的联系：任意有界数列都有收敛子列；若只出现有限多个不同值，至少一个值出现无限次，取常值子列；若出现无限多个值，使用集合版本和递增下标挑选即可。

22 原课件第 22 页：二分立方体证明：不断保留无限多个点

Bolzano-Weierstrass Theorem

Proof. Note that E is a bounded infinite set, there exists a constant M such that, for any $x = (x_1, \cdots, x_N) \in E$, $|x_n| \leqslant M$ for $n = 1, \cdots, N$, and hence E includes in N-dimensional cube

$$R_0 = E[(x_1, \cdots, x_N); |x_n| \leqslant M \text{ for } n = 1, \cdots, N].$$

We use axises to divide R_0 into 2^N small closed cubes, then at least one small closed cube includes infinite points in E (otherwise E is a finite set!), we call such a small closed cube R_1 with edge M. Generally, we assume that there exists a sequence of closed cubes whose edges are parallel to axises.

$$R_0 \supseteq R_1 \supseteq \cdots \supseteq R_N, \quad R_N \cap E \text{ is an infinite set,}$$
$$\text{the edge of } R_n \text{ is } \frac{M}{2^{n-1}} \text{ for } n = 1, 2, \cdots, N$$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

把有界集 E 放进 $[-M, M]^N$。沿每个坐标二分，得到 2^N 个闭小方块。若每块只含有限个 E 点，则有限块合起来仍只有有限点，与 E 无限矛盾。因此至少一块仍含无限个点。选这一块再二分，如此构造

$$C_0 \supseteq C_1 \supseteq \cdots, \qquad \text{边长}(C_k) = 2M/2^k, \quad |C_k \cap E| = \infty.$$

小方块可以在边界上相交，不影响“有限个有限集之并仍有限”的推理。

23 原课件第 23 页：二分证明的收尾：共同点就是聚点

Bolzano-Weierstrass Theorem

We use lines parallel to axises to divide R_N into 2^N small closed cubes, then at least one small closed cube includes infinite points in E (otherwise E is a finite set!), we call such a small closed cube R_{N+1} with edge $\frac{M}{2^N}$. Thus we obtain a sequence of decreasing closed cubes whose edges are parallel to axises.

$$R_0 \supseteq R_1 \supseteq \cdots \supseteq R_n \supseteq \cdots, \quad R_n \cap E \text{ is an infinite set,}$$
$$\text{and the edge of } R_n \text{ is } \frac{M}{2^{n-1}} \text{ for } n = 1, 2, \cdots$$

Note $\lim\limits_{n\to\infty} \frac{M}{2^{n-1}} = 0$, then, from Cantor Nested Intervals Theorem, there exists a unique point $P \in \bigcap\limits_{n=1}^{\infty} R_n$. We now check that this P is one of accumulation points of E.

Define $N(P, \delta)$ as any neighborhood of center P. Note $\lim\limits_{n\to\infty} \frac{M}{2^{n-1}} = 0$, then for enough large natural number n_0, $\frac{M}{2^{n_0}} < \frac{\delta}{n_0^{\frac{1}{2}}}$, $n_0^{\frac{1}{2}} \frac{M}{2^{n_0}} < \delta$,. The diagonal length of R_{n_0+1} is strictly less than δ, then $R_{n_0+1} \subseteq N(P, \delta)$, and hence $E \cap R_{n_0+1} \subseteq E \cap N(P, \delta)$. There exist infinite points of $N(P, \delta)$ in E.

Q.E.D.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

每个 C_k 是 N 个闭区间的乘积；逐坐标应用闭区间套定理，得到唯一共同点 x。给定 $r > 0$，选 k 使 $\sqrt{N}(2M/2^k) < r$，则 C_k 的欧氏直径小于 r。因为 $x \in C_k$，故 $C_k \subset B_r(x)$。而 $C_k \cap E$ 无限，所以每个 $B_r(x)$ 都包含无限个 E 点，x 是聚点。 **课件修正：**对角线系数应为固定维数的 $\sqrt{N}$，不能误写成不断变化的子块编号的平方根。最大范数下直径直接等于边长。

24 原课件第 24 页：孤立点与可数性的证明起点

Bolzano-Weierstrass Theorem

In mathematics, a point P is called an isolated point of a set E if P is an element of E but there exists a neighborhood of P which does not contain any other points of E.

Point P is an isolated point of set E if $P \in E$ and $P \notin D(P, \delta)$ for $\delta > 0$.

The boundary point which is not accumulation point is called to be an isolated point. The isolated point belongs to E. A set E is isolated if every point is isolated in E.

Each isolated set is a countable (finite or denumerable) set. In fact, if E is an isolated set, then, for any $P \in E$, there exists $\delta_P > 0$ such that $N(P, 2\delta_P) \cap E = \{P\}$. We pick a rational point R_P in $N(P, \delta_P)$. If $P_1 \neq P_2$, then $R_{P_1} \neq R_{P_2}$. Otherwise, $R_{P_1} = R_{P_2}$, then

$$\begin{aligned} \rho(P_1, P_2) &\leqslant \rho(P_1, R_{P_1}) + \rho(R_{P_1}, P_2) = \rho(P_1, R_{P_1}) + \rho(P_2, R_{P_2}) \\ &< \delta_{P_1} + \delta_{P_2} \leqslant 2 \max\{\delta_{P_1}, \delta_{P_2}\}. \end{aligned}$$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

x 是 E 的孤立点，正确条件是 $x \in E$ 且存在 $r > 0$ 使 $B_r(x) \cap E = \{x\}$，等价于去心球与 E 无交。**课件修正：** $x \notin D(x, r)$ 总是成立，单独这一句不能定义孤立点。若所有点都孤立，对每个 $x \in E$ 选 $r_x > 0$ 使 $B_{2r_x}(x) \cap E = \{x\}$，再在 $B_{r_x}(x)$ 中选一个有理坐标点 $q_x \in \mathbb{Q}^N$。有理坐标点的存在来自每一坐标上有理数稠密；下一页证明这些 q_x 不会重复。

25 原课件第 25 页：把孤立点注入有理坐标点

Bolzano-Weierstrass Theorem

(1) If $\max\{\delta_{P_1}, \delta_{P_2}\} = \delta_{P_1}$, then $\rho(P_1, P_2) < 2\delta_{P_1}$, then $P_2 \in N(P_1, 2\delta_{P_1})$ and $N(P_1, 2\delta_{P_1}) \cap E = \{P_1, P_2\}$.

(2) If $\max\{\delta_{P_1}, \delta_{P_2}\} = \delta_{P_2}$, then $\rho(P_1, P_2) < 2\delta_{P_2}$, then $P_1 \in N(P_2, 2\delta_{P_2})$ and $N(P_2, 2\delta_{P_2}) \cap E = \{P_1, P_2\}$.

Therefore there exists an one - to -one correspondence from $P \in E$ to R_P. Note that the set of all rationals is denumerable, then E is a countable (finite or denumerable) set.

E is an isolated set if and only if $E \cap E' = \varnothing$.

E is a discrete set if $E' = \varnothing$. Any discrete set is an isolated set, but an isolated set may not be a discrete set. For example, the point set $\left\{\frac{1}{n}; n = 1, 2, \cdots\right\} \subseteq \mathbb{R}$ is an isolated set, but it is not a discrete set because 0 is an accumulation point.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

若不同的 x, y 选中了同一个 q，则 $\|x - y\| < r_x + r_y \le 2\max(r_x, r_y)$。若最大者是 r_x，则 $y \in B_{2r_x}(x) \cap E$，与该交集只有 x 矛盾；另一情形同理。故 $x \mapsto q_x$ 是单射。$\mathbb{Q}^N$ 是有限个可数集之积，仍可数，所以 E 可数。本课把 $E \cap E' = \varnothing$ 称为所有点孤立，把 $E' = \varnothing$ 称为 discrete。其他教材常把前者也称为离散子空间，答题应先写条件再写术语。$\{1/k\}$ 的每一点都孤立，但0仍是外部聚点。

26 原课件第 26 页：聚点也可以用去心邻域定义

Accumulation Points, Interior Points, and Boundary Points

Theorem 7. Let E be a set of $\mathbb{R}^N$ and P_0 be a given point in $\mathbb{R}^N$. $P_0 \in E'$ if and only if, in any neighborhood $N(P, \delta)$ of point P_0, there exists a point P', differing from P_0, in E (there exist infinite such points P').

Proof. [Necessity] If $P_0 \in E'$, then, in any neighborhood $N(P, \delta)$ of point P_0, there exists a point P', differing from P_0, in E (there exist infinite such points P').

If $P_0 \in E'$, then for any $\delta > 0$ there exist infinite many points in $N(P_0, \delta) \cap E$. Take $P \in N(P_0, \delta)$, then $0 \leqslant d = \rho(P, P_0) < \delta$. Thus for any $P' \in N(P_0, \delta - d)$ we have
$\rho(P, P') \leqslant \rho(P, P_0) + \rho(P_0, P') < d + \delta - d = \delta$, and hence
$N(P_0, \delta - d) \subseteq N(P, \delta)$ and $N(P_0, \delta - d) \cap E \subseteq N(P, \delta) \cap E$.

Note that $N(P_0, \delta - d) \cap E$ has infinite many points, then there exists a point P', differing from P_0, in $N(P_0, \delta - d) \cap E \subseteq N(P, \delta) \cap E$ (there exist infinite such points P').

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

等价条件是 $x \in E' \iff \forall r > 0,\ (B_r(x) \setminus \{x\}) \cap E \neq \varnothing$。正向显然：无限多个点不可能只有中心一个。若课件说一个“含 x 的邻域”却以 y 为中心，只要 $x \in B_r(y)$，仍可找到以 x 为中心的小球 $B_{r-\|x-y\|}(x) \subset B_r(y)$。因为半径差严格为正，两种邻域表述等价。

27 原课件第 27 页：一个额外点为什么能推出无限多个点

Accumulation Points, Interior Points, and Boundary Points

[Sufficiency] If, in any neighborhood $N(P,\delta)$ of point P_0, there exists a point P', differing from P_0, in E (there exist infinite such points P'), then $P_0 \in E'$.

In any neighborhood $N(P,\delta)$ of point P_0, there exists a point P_1, differing from P_0, in E. Write $0 < \delta_1 = \rho(P_1, P_0) < \delta$. In δ_1 neighborhood $N(P_0,\delta_1)$ of center P_0, there exists a point P_2, differing from P_0, in E. Write $0 < \delta_2 = \rho(P_2, P_0) < \delta_1$, then $P_2 \in N(P_0,\delta_1)$. Generally, in δ_n neighborhood $N(P_0,\delta_n)$ of center P_0, there exists a point P_{n+1}, differing from P_0, in E. Write $0 < \delta_{n+1} = \rho(P_{n+1}, P_0) < \delta_n$, then $P_{n+1} \in N(P_0,\delta_n)$. From induction, there exist $\{\delta_n : n \in \mathbb{N}\}$ with $\delta_{n+1} < \delta_n$ for $n \in \mathbb{N}$ such that point P_{n+1}, differing from P_0, in E. Write $0 < \delta_{n+1} = \rho(P_{n+1}, P_0) < \delta_n$, then $P_{n+1} \in N(P_0,\delta_n)$. Therefore there exist infinitely many points in $N(P_0,\delta) \cap E$, and hence $P_0 \in E'$. Q.E.D.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

逆向假设每个去心球都碰到 E。若某个 $B_r(x)$ 只含有限个异于 x 的点 $y_1, \ldots, y_m$，取 $0 < s < \min(r, \|y_1 - x\|, \ldots, \|y_m - x\|)$，则去心球 $B_s(x) \setminus \{x\}$ 不碰 E，矛盾。若原球根本没有异点，已经直接矛盾。因此每个球包含无限点。 课件递归选距离严格下降的点也能证明点数无限；但“严格下降的正数列”并不自动趋0，例如 $1+1/k$。若要同时构造收敛列，应额外把第 k 次半径限制到 $1/k$。

28 原课件第 28 页：邻域换中心与内部的局部稳定性

Accumulation Points, Interior Points, and Boundary Points

Theorem 8. Let E be a set of $\mathbb{R}^N$ and P_0 be a given point in $\mathbb{R}^N$. $P_0 \in E^\circ$ if and only if there exists a neighborhood $N(P,\delta)$ of point P_0 such that $N(P,\delta) \subseteq E$.

Proof. [Necessity] If $P_0 \in E^\circ$, then there exists a neighborhood $N(P,\delta)$ of point P_0 such that $N(P,\delta) \subseteq E$.
If $P_0 \in E^\circ$, then there exists $\delta > 0$ such that $N(P_0,\delta) \subseteq E$. Take $P \in N(P_0,\delta)$, then $0 \leqslant d = \rho(P,P_0) < \delta$. Thus for any $P' \in N(P,\delta-d)$ we have
$\rho(P_0,P') \leqslant \rho(P,P_0) + \rho(P,P') < d + \delta - d = \delta$, and hence $P' \in N(P_0,\delta)$. Thus $N(P,\delta-d) \subseteq N(P_0,\delta)$ and $N(P,\delta-d) \subseteq E$.

[Sufficiency] If there exists a neighborhood $N(P,\delta)$ of point P_0 such that $N(P,\delta) \subseteq E$, then $P_0 \in E^\circ$.
There exists a neighborhood $N(P,\delta)$ of point P_0 such that $N(P,\delta) \subseteq E$, then $N(P_0,\delta-\rho(P_0,P)) \subseteq N(P,\delta) \subseteq E$, and hence $P_0 \in E^\circ$. Q.E.D.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

若 $B_r(x) \subseteq E$，且 $y \in B_r(x)$，记 $d = \|x-y\| < r$。对 $z \in B_{r-d}(y)$，三角不等式给 $\|z-x\| < r-d+d = r$，故 $z \in E$。所以原球中的每个点也是内点。这一小计算将反复出现：可移动的余量等于原半径减去已走过的距离。若从一个非中心邻域反推中心邻域，也用同样的余量公式。

29 原课件第 29 页：闭集的定义与三个基本例子

Closed Set

Let $E \subseteq \mathbb{R}^N$. If $E' \subseteq E$, all accumulation points belong to E, then E is a closed set (here, the empty set is defined as a closed set). Note $\overline{E} = E \cup E'$, we say $\overline{E}$ is the closure E (E is a closed set if and only if $E = \overline{E}$).

Example 1. The closed rectangular in $\mathbb{R}^N$ is a closed set. $\mathbb{R}^N$ itself is a closed set. The closure of the rational number set $\mathbb{Q}$ is $\mathbb{R}$.

Example 2. If $f(x)$ is defined on $\mathbb{R}^N$, then the sufficient and necessary conditions for $f \in C\left(\mathbb{R}^N\right)$ are, for any $t \in \mathbb{R}$, the sets

$$E_1 = \left\{x \in \mathbb{R}^N \mid f(x) \geqslant t\right\} \qquad \text{and} \qquad E_2 = \left\{x \in \mathbb{R}^N \mid f(x) \leqslant t\right\}$$

are closed sets.

Example 3. The closure of the open ball $N(x_0, r)$ in $\mathbb{R}^N$ is the closed ball $B(x_0, r)$:

$$B(x_0, r) = \overline{N(x_0, r)} = \left\{x \in \mathbb{R}^N \mid |x - x_0| \leqslant r\right\}.$$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

E 闭是指 $E' \subseteq E$，等价于 $\overline{E} = E$。更适合考试的版本：只要 $x_k \in E$ 且 $x_k \to x$，就有 $x \in E$。证明这个版本时，若极限就是某项值则已经在 E；否则可抽出趋向 x 的不同点，故 $x \in E'$。闭长方体的每个坐标不等式在取极限后保留，所以闭；$\mathbb{R}^N$ 和空集也闭。有理数稠密说明 $\overline{\mathbb{Q}} = \mathbb{R}$，因此 $\mathbb{Q}$ 不闭。连续函数上下水平集的判别在下一页证明；开球闭包等于闭球要求半径正，见第31页。

30 原课件第 30 页：连续性与上下水平集闭：补上子列步骤

Closed Set

Example 2. If $f(x)$ is defined on $\mathbb{R}^N$, then the sufficient and necessary conditions for $f \in C\left(\mathbb{R}^N\right)$ are, for any $t \in \mathbb{R}$, the sets

$$E_1 = \left\{x \in \mathbb{R}^N \mid f(x) \geqslant t\right\} \qquad \text{and} \qquad E_2 = \left\{x \in \mathbb{R}^N \mid f(x) \leqslant t\right\}$$

are closed sets.

Proof [Necessity] For E_1, if $\{x_k; k \in \mathbb{N}\} \subseteq E_1$ and $\lim\limits_{k\to\infty} x_k = x_0$, from the continuity of $f(x)$ and $f(x_k) \geqslant t$, then $f(x_0) = \lim\limits_{k\to\infty} f(x_k) \geqslant t$, i.e. $x_0 \in E_1$. For E_2, if $\{x_k; k \in \mathbb{N}\} \subseteq E_2$ and $\lim\limits_{k\to\infty} x_k = x_0$, from the continuity of $f(x)$ and $f(x_k) \leqslant t$, then $f(x_0) = \lim\limits_{k\to\infty} f(x_k) \leqslant t$, i.e. $x_0 \in E_2$.

[Sufficiency] We adopt the reduction to absurdity. We assume that there exists $x_0 \in \mathbb{R}^N$ which is not a continuous point of $f(x)$, then there is $\varepsilon_0 > 0$ and point sequence $\{x_k; k \in \mathbb{N}\}$ such that $\lim\limits_{k\to\infty} x_k = x_0$. For every $k \in \mathbb{N}$,

$$f(x_k) \leqslant f(x_0) - \varepsilon_0 \qquad \text{or} \qquad f(x_k) \geqslant f(x_0) + \varepsilon_0.$$

If $f(x_k) \leqslant f(x_0) - \varepsilon_0$, then $x_k \in E_2[t = f(x_0) - \varepsilon_0]$, but $x_0 \notin E_2[t = f(x_0) - \varepsilon_0]$, which is a contradiction! If $f(x_k) \geqslant f(x_0) + \varepsilon_0$, then $x_k \in E_1[t = f(x_0) + \varepsilon_0]$, but $x_0 \notin E_1[t = f(x_0) + \varepsilon_0]$, which is a contradiction!

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

正向 若 f 连续，且 $x_k \to x$、$f(x_k) \geq t$，则 $f(x) = \lim f(x_k) \geq t$，故 $\{f \geq t\}$ 闭；$\{f \leq t\}$ 同理。

反向 若在 x 不连续，则存在 $\varepsilon_0 > 0$，对每个 k 都能选 $\|x_k - x\| < 1/k$ 而 $|f(x_k) - f(x)| \geq \varepsilon_0$。每项至少属于以下一类：$f(x_k) \leq f(x) - \varepsilon_0$ 或 $f(x_k) \geq f(x) + \varepsilon_0$。两类中至少一类有无限多个下标，抽取该类子列。例如第一类得到闭集 $\{f \leq f(x) - \varepsilon_0\}$ 中的数列趋于 x，但 x 不在该闭集，矛盾。第二类同理。 **关键补足：** 不能假定所有项天然落在同一边；必须先用无限鸽巢原理抽子列。

31 原课件第 31 页：开球的闭包与稠密

Closed Set

Example 3. The closure of the open ball $N(x_0, r)$ in $\mathbb{R}^N$ is the closed ball $B(x_0, r)$:

$$B(x_0, r) = \overline{N(x_0, r)} = \left\{x \in \mathbb{R}^N \mid |x - x_0| \leqslant r\right\}.$$

Proof Define $F = \left\{x \in \mathbb{R}^N \mid |x - x_0| \leqslant r\right\}$, then f is a closed set, so $\overline{N(x_0, r)} \subseteq F$. Conversely, if $x \in F$, define $x_k = \frac{1}{k}x_0 + \left(1 - \frac{1}{k}\right)x$, and

$$|x_0 - x_k| = \left(1 - \frac{1}{k}\right)|x - x_0| \leqslant \left(1 - \frac{1}{k}\right) r < r \text{ and } |x - x_k| = \frac{1}{k}|x - x_0| \leqslant \frac{1}{k} r < r.$$

This indicates that $\{x_k; k \in \mathbb{N}\} \subseteq N(x_0, r)$, and $\lim\limits_{k\to\infty} x_k = x$. Thus $x \in \overline{N(x_0, r)}$, that is, $F \subseteq \overline{N(x_0, r)}$.

Note: If $A \subseteq S$ and $\overline{A} = S$, then A is said to be dense in S, or A is a dense subset of S.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

设 $r > 0$。若 $x_k \in B_r(a)$ 且 $x_k \to x$，距离连续性给 $\|x - a\| \leq r$，故 $\overline{B_r(a)} \subseteq \overline{B}_r(a)$。反向对 $x \in \overline{B}_r(a)$，取 $x_k = a + (1 - 1/k)(x - a)$，则 $\|x_k - a\| = (1 - 1/k)\|x - a\| < r$，且 $x_k \to x$。因此等号成立。A 在 S 中稠密，准确表述是 $A \subseteq S$ 且 $S \subseteq \overline{A}$，或相对闭包 $\overline{A}^S = S$。若直接写环境空间中的 $\overline{A} = S$，还暗中要求 S 闭。后面的第84–86页使用前一种一般定义。

32 原课件第 32 页：抽象闭包运算：四条规则



If with each subset X of a set S we associate a subset $\overline{X}$ of S such that

(1) $X \subseteq \overline{X}$,

(2) $X_1 \subseteq X_2 \Longrightarrow \overline{X}_1 \subseteq \overline{X}_2$,

(3) $\overline{\overline{X}} = \overline{X}$,

(4) $\overline{\varnothing} = \varnothing$,

then we call the correspondence $X \to \overline{X}$ a closure operation.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

课件列出：扩大集合、保持包含、做两次等于一次、空集仍为空集。把运算记为 c，就是 $A \subseteq c(A)$，$A \subseteq B \Rightarrow c(A) \subseteq c(B)$，$c(c(A)) = c(A)$，$c(\varnothing) = \varnothing$。这些规则描述抽象闭包算子。**边界提醒：**仅这四条还不能保证通常拓扑闭包的有限并公式；拓扑闭包还满足 $c(A \cup B) = c(A) \cup c(B)$。本讲的欧氏空间闭包满足更强的这些性质，抽象段落不要无条件套用全部欧氏结论。

33 原课件第 33 页：导集与闭包本身都是闭集

Closed Set

Theorem 1. E' and $\overline{E}$ are closed sets.

Proof. If $P_0 \in (E')'$, then, in any neighborhood $N(P, \delta)$ of point P_0, there exists a point $P_1 \in E'$ with $P_1 \neq P_0$ from Theorem 7 in Subsection 2, and hence, in any neighborhood $N(P', \delta')$ of point P_1, there exists a point $P_2 \in E$ with $P_2 \neq P_1$ and $P_2 \neq P_0$ from Theorem 7 in Subsection 2, thus $P_0 \in E'$. Therefore E' is a closed set.

Note $\overline{E} = E \cup E'$, then $\overline{E}' = E' \cup (E')' \subseteq E' \cup E' = E' \subseteq \overline{E}$, and hence $\overline{E}$ is a closed set. Q.E.D.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

证明 E' 闭：设 $x \in (E')'$。给定 $r > 0$，选 $y \in E'$ 满足 $0 < \|y - x\| < r/2$，再取 $s < \min(r - \|y - x\|, \|y - x\|)$。由于 y 是 E 的聚点，可选 $z \in E \cap B_s(y)$。这个小球不含 x，又包含于 $B_r(x)$，因此找到了 E 中异于 x 的近点。由第26–27页，$x \in E'$。再用有限并导集公式：$(\overline{E})' = (E \cup E')' = E' \cup (E')' \subseteq E' \subseteq \overline{E}$，所以 $\overline{E}$ 闭。课件略去的小球半径选择正是证明严密性所在。

34 原课件第 34 页：有限个闭集之并仍闭：导集方法

Closed Set

Theorem 2. The union of finite closed sets is a closed set.

Proof. Define $F = \bigcup_{n=1}^{N} F_n$ where F_n is a closed set for $n = 1, \cdots, N$. Note that F_n is a closed set, $F_n' \subseteq F_n$, then

$$F' = \left(\bigcup_{n=1}^{N} F_n\right)' = \bigcup_{n=1}^{N} F_n' \subseteq \bigcup_{n=1}^{N} F_n = F.$$

Therefore F is a closed set. Q.E.D.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

设 $F = F_1 \cup \cdots \cup F_m$，每个 F_i 闭。由第19页的有限并公式，$F' = F_1' \cup \cdots \cup F_m' \subseteq F_1 \cup \cdots \cup F_m = F$，于是 F 闭。注意“finite closed sets”在这里表示有限个闭集，不是每个闭集本身只有有限个点。考试写明 $m < \infty$，防止把定理误用到无限并。

35 原课件第 35 页：同一个有限并结论：闭包方法

Closed Set

Theorem 2. The union of finite closed sets is a closed set.

Proof. Define $F = \bigcup_{n=1}^{N} F_n$ where F_n is a closed set for $n = 1, \cdots, N$. Note that F_n is a closed set, $\overline{F}_n = F_n \cup F_n'$, then

$$\begin{aligned}
\overline{F} &= \overline{\bigcup_{n=1}^{N} F_n} = \left(\bigcup_{n=1}^{N} F_n\right) \cup \left(\bigcup_{n=1}^{N} F_n\right)' \\
&= \left(\bigcup_{n=1}^{N} F_n\right) \cup \left(\bigcup_{n=1}^{N} F_n'\right) \\
&= \bigcup_{n=1}^{N} \left(F_n \cup F_n'\right) = \bigcup_{n=1}^{N} \overline{F}_n = \bigcup_{n=1}^{N} F_n = F.
\end{aligned}$$

Therefore F is a closed set. Q.E.D.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页给另一种代数写法：

$$\overline{\bigcup_{i=1}^{m} F_i} = \bigcup_{i=1}^{m} (F_i \cup F_i') = \bigcup_{i=1}^{m} \overline{F_i} = \bigcup_{i=1}^{m} F_i.$$

第一个等号用闭包等于本集并导集及有限并导集公式；最后一个等号用每个 F_i 闭。这与上一页是一条证明的两种记号。复习时应能解释每个等号来自哪条定义，不能仅抄下一串等号。

36 原课件第 36 页：可数个闭集之交仍闭：导集方法

Closed Set

Theorem 3. The intersection of multiple closed sets is a closed set.

Proof. Define $F = \bigcap_{n=1}^{\infty} F_n$ where F_n is a closed set for $n = 1, 2, \cdots$.

Note $F \subseteq F_n$, then $F' \subseteq F_n'$. Thus $F' \subseteq \bigcap_{n=1}^{\infty} F_n'$. Note that F_n is a closed set, $F_n' \subseteq F_n$, then

$$\left(\bigcap_{n=1}^{\infty} F_n\right)' = F' \subseteq \bigcap_{n=1}^{\infty} F_n' \subseteq \bigcap_{n=1}^{\infty} F_n = F.$$

Therefore F is a closed set. Q.E.D.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

设 $F = \bigcap_{k=1}^{\infty} F_k$。因为 $F \subseteq F_k$，单调性给 $F' \subseteq F_k' \subseteq F_k$，对每个 k 都成立，于是 $F' \subseteq \bigcap_k F_k = F$。这里不用交换“求导集”与“无限交”：一般只有包含，不应写成等号。这个证明实际上完全没有用到可数性，下一组页面将推广到任意指标集。

37 原课件第 37 页：可数闭交的闭包写法

Closed Set

Theorem 3. The intersection of multiple closed sets is a closed set.

Proof. Define $F = \bigcap_{n=1}^{\infty} F_n$ where F_n is a closed set for $n = 1, 2, \cdots$.

Note $F \subseteq F_n$, then $\overline{F} \subseteq \overline{F}_n$. Thus $\overline{F} \subseteq \bigcap_{n=1}^{\infty} \overline{F}_n$. Note that F_n is a closed set, $\overline{F}_n = F_n$, then

$$\overline{\bigcap_{n=1}^{\infty} F_n} = \overline{F} \subseteq \bigcap_{n=1}^{\infty} \overline{F}_n = \bigcap_{n=1}^{\infty} F_n = F.$$

Therefore F is a closed set. Q.E.D.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

对每个 k 有 $\overline{F} \subseteq \overline{F_k} = F_k$，所以 $\overline{F} \subseteq F$。再结合总成立的 $F \subseteq \overline{F}$，得到 $F = \overline{F}$。本页比上一页多用一次“双向包含推出相等”的动作。若 F 为空，也一样成立，不需要单独假设交集非空。

38 原课件第 38 页：任意闭集族的交：不需要数得出来

Closed Set

Theorem 3. If $\{F_\gamma : \gamma \in \Gamma\}$ is a family of closed sets in $\mathbb{R}^N$, then its intersection $F = \bigcap_{\gamma\in\Gamma} F_\gamma$ is a closed set.

Proof. Define $F = \bigcap_{\gamma\in\Gamma} F_\gamma$ where F_γ is a closed set for $\gamma \in \Gamma$. Note $F \subseteq F_\gamma$, then $F' \subseteq F'_\gamma$. Thus $F' \subseteq \bigcap_{\gamma\in\Gamma} F'_\gamma$. Note that F_γ is a closed set, $F'_\gamma \subseteq F_\gamma$, then

$$\left(\bigcap_{n=1}^{\infty} F_n\right)' = F' \subseteq \bigcap_{\gamma\in\Gamma} F'_\gamma \subseteq \bigcap_{\gamma\in\Gamma} F_\gamma = F.$$

Therefore F is a closed set. Q.E.D.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

设 Γ 可以不可数，$F = \bigcap_{\gamma\in\Gamma} F_\gamma$。任取 $x \in F'$，因为 $F \subseteq F_\gamma$，可知 $x \in F'_\gamma \subseteq F_\gamma$，对所有 γ 成立，所以 $x \in F$。原页公式残留可数指标 n 的位置应统一改为 $\gamma \in \Gamma$。若指标集为空，约定交集为整个环境空间，仍为闭集。

39 原课件第 39 页：任意闭交的闭包证明与笔误

Closed Set

Theorem 3. If $\{F_\gamma : \gamma \in \Gamma\}$ is a family of closed sets in $\mathbb{R}^N$, then its intersection $F = \bigcap_{\gamma\in\Gamma} F_\gamma$ is a closed set.

Proof. Define $F = \bigcap_{\gamma\in\Gamma} F_\gamma$ where F_γ is a closed set for $\gamma \in \Gamma$. Note $F \subseteq F_\gamma$, then $\overline{F} \subseteq \overline{F}_\gamma$. Thus $\overline{F} \subseteq \bigcap_{\gamma\in\Gamma} \overline{F}_\gamma$. Note that F_γ is a closed set, $\overline{F}_\gamma \subseteq \overline{F}$, then

$$\overline{\bigcap_{n=1}^{\infty} F_n} = \overline{F} \subseteq \bigcap_{\gamma\in\Gamma} \overline{F}_\gamma = \bigcap_{\gamma\in\Gamma} F_\gamma = F.$$

Therefore F is a closed set. Q.E.D.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

延续第37页，把 k 换成 γ：$\overline{F} \subseteq \overline{F_\gamma} = F_\gamma$，故 $\overline{F} \subseteq \bigcap_\gamma F_\gamma = F$。**课件修正：**每个 F_γ 闭意味着 $\overline{F_\gamma} = F_\gamma$；不意味着 $\overline{F_\gamma} \subseteq F$，因为交集通常比单个集合小。不要把单调性方向倒过来。

40 原课件第 40 页：为什么无限并会出现新极限点

Closed Set

Note that the union of an infinite number of closed sets is not necessarily a closed set. For example, let

$$F_k = \left[\frac{1}{k+1}, \frac{1}{k}\right] \subseteq \mathbb{R} \qquad (k = 1, 2, \cdots),$$

then $\bigcup_{k=1}^{\infty} F_k = (0, 1]$. Furthermore, $\overline{\bigcup_{k=1}^{\infty} F_k} \neq \bigcup_{k=1}^{\infty} \overline{F}_k$.
But we have the following simple facts: Let $E_\gamma \subseteq \mathbb{R}^N$ for $\gamma \in \Gamma$, then

$$\bigcup_{\gamma\in\Gamma} \overline{E}_\gamma \subseteq \overline{\bigcup_{\gamma\in\Gamma} E_\gamma} \qquad \text{and} \qquad \overline{\bigcap_{\gamma\in\Gamma} E_\gamma} \subseteq \bigcap_{\gamma\in\Gamma} \overline{E}_\gamma.$$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$F_k = [1/(k+1), 1/k]$ 都闭，但 $\bigcup_k F_k = (0, 1]$ 不含聚点0，所以不闭。具体看 $x_k = 1/k$：每项落在某个闭区间，但没有一个固定的区间包含无限多个趋0的项。 一般总有

$$\bigcup_{\gamma} \overline{E_\gamma} \subseteq \overline{\bigcup_{\gamma} E_\gamma}, \qquad \overline{\bigcap_{\gamma} E_\gamma} \subseteq \bigcap_{\gamma} \overline{E_\gamma}.$$

第一式对每个 $E_\gamma \subseteq \bigcup E_\gamma$ 用单调性再取并；第二式对 $\bigcap E_\gamma \subseteq E_\gamma$ 用单调性再取交。等号一般不成立。

41 原课件第 41 页：闭包是包含原集合的最小闭集



Theorem 4. The closure $\overline{A}$ of A is the intersection of all the closed sets which contain A.

Proof. Let E be the intersection of all the closed sets which contain A; by the preceding theorem, E is closed. Since $\overline{A}$ is a closed set which contains A, we have $E \subseteq \overline{A}$. But we also have $E = \overline{E} \supseteq \overline{A}$, by condition (2), whence $E = \overline{A}$. Q.E.D.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

令 $\mathcal{F} = \{F : F \text{ 闭且 } A \subseteq F\}$，$C = \bigcap_{F \in \mathcal{F}} F$。由于 $\overline{A}$ 本身闭且包含 A，它是被求交的成员之一，故 $C \subseteq \overline{A}$。另一方面每个 $F \in \mathcal{F}$ 满足 $\overline{A} \subseteq \overline{F} = F$，所以 $\overline{A} \subseteq C$。两边合并得 $C = \overline{A}$。“最小”按集合包含比较，不是按元素数量比较；$\mathcal{F}$ 至少含有整个空间，因此并非无从取交。

42 原课件第 42 页：递减非空紧集的交非空：简洁完整证明

Closed Set

Theorem 5 (Cantor Nested Closed Sets Theorem) If $\{F_k; k \in \mathbb{N}\}$ is a sequence of non-empty bounded closed sets in $\mathbb{R}^N$ with $F_1 \supseteq F_2 \supseteq \cdots \supseteq F_k \supseteq \cdots$, then

$$\bigcap_{k=1}^{\infty} F_k \neq \varnothing.$$

Proof. If there are infinitely many identical sets in $\{F_k; k \in \mathbb{N}\}$, then there is a natural number $K \in \mathbb{N}$, define $F_k = F_K$ for $k \geqslant K$.

$$\bigcap_{k=1}^{\infty} = F_K \neq \varnothing.$$

Now let's assume that for all $k \in \mathbb{N}$, $F_{K+1} \subseteq F_k$, and F_{K+1} is the proper subset of F_k, i.e. $F_k \setminus F_{K+1} \neq \varnothing$ for $k \in \mathbb{N}$. We choose $x_k \in F_k \setminus F_{K+1}$ for $k \in \mathbb{N}$, then $\{x_k; k \in \mathbb{N}\}$ is a sequence of bounded distinct points in $\mathbb{R}^N$. According to Bolzano-Weierstrass theorem, there are $\{x_{k_q}; q \in \mathbb{N}\}$ and $x \in \mathbb{R}^N$ such that

$$\lim_{q \to \infty} |x_{k_q} - x| = 0.$$

Since each F_k is a closed set, then $x \in F_k$ for $k \in \mathbb{N}$, i.e., $x \in \bigcap_{k=1}^{\infty} F_k$.

Q.E.D.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

设 $F_1 \supseteq F_2 \supseteq \cdots$，各集非空、闭且 F_1 有界。从每个 F_k 任取 $x_k \in F_k$。整个数列都在 F_1 中，故有收敛子列 $x_{k_j} \to x$。任意固定 m，当 $k_j \geq m$ 时，$x_{k_j} \in F_{k_j} \subseteq F_m$。由 F_m 闭得到 $x \in F_m$。因为 m 任意，$x \in \bigcap_m F_m$。这个证明自动处理重复集合、重复点，不必按课件分情况。交集未必只有一点；加上直径趋0才唯一。若去掉有界性，$F_k = [k, \infty)$ 给出空交；去掉闭性，$F_k = (0, 1/k)$ 也给出空交。

43 原课件第 43 页：闭集习题详解

Closed Set

Questions. Answer the following questions:

1. Let $E \subseteq \mathbb{R}$ be a non-empty set. If any subset of E is a closed set, is E a finite set?
2. Let E_1 and E_2 be the sets of $\mathbb{R}$, whether must the equation $\overline{E_1 \cap E_2} = \overline{E}_1 \cap \overline{E}_2$ hold?
3. Let $E_k \subseteq \mathbb{R}^N$ for $k \in \mathbb{N}$, define $E = \bigcup_{k=1}^{\infty} E_k$. If $x_0 \in E'$, whether must there be E_{k_0} such that $x_0 \in E'_{k_0}$?
4. Let $E \subseteq \mathbb{R}^N$. Prove that $\overline{E}$ is the intersection of all closed sets F containing E: $\overline{E} = \bigcap_{F \supseteq E} F$.
5. Let $F \subseteq \mathbb{R}$ be a bounded closed set and $f(x)$ be a (real-valued) function defined on F. If for any $x_0 \in F'$, $f(x) \to +\infty$ as $x \to x_0$ and $x \in F$, prove that F is a countable set.
6. Let $f \in C(\mathbb{R})$, prove that $F = \{(x, y), f(x) \geqslant y\}$ is a closed set in $\mathbb{R}^2$.
7. Try to list countable collection of disjoint dense countable sets in $\mathbb{R}$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

1. 不必有限：$\mathbb{Z}$ 的任意子集都无有限聚点，故在 $\mathbb{R}$ 中闭。2. 不成立：$E_1 = (0,1), E_2 = (1,2)$，交为空而闭包交含1。3. 不成立：$E_k = \{1/k\}$，每个导集为空，但并集的导集含0。4. 闭包是闭的且含 E，任何闭集含 E 也含其全部极限点，故恰为交。5. 对每个整数 m 置 $A_m = \{x \in F : f(x) \leq m\}$。若 A_m 无限，紧性给一列互异点收敛到 $x_0 \in F'$，与沿这些点函数趋于无穷矛盾。故每个 A_m 有限，$F = \bigcup_m A_m$ 可数。6. 若 $(x_n, y_n) \to (x, y)$ 且 $f(x_n) \geq y_n$，连续性和不等式保极限给 $f(x) \geq y$。7. 取 $\mathbb{Q} + n\sqrt{2}$（$n \in \mathbb{Z}$）：各集合稠密可数，若相交则 $(n-m)\sqrt{2} \in \mathbb{Q}$，推出 $n = m$。

44 原课件第 44 页：内点与内部

Open Set

Region is a special point set in mathematical analysis. An important proposition for region D is that, if point $P \in D$, then there exists a small area including P in D. Thus every point in D is an interior point.

If $E^\circ = E$, every point in E is interior, then E is an open set.

Theorem: If G is a non-empty set in $\mathbb{R}^N$, then G is an open set if and only if, for any point $x \in G$, there is $\delta > 0$ such that $B(x, \delta) \subseteq G$

The unit open interval $(0, 1)$ is an open set in $\mathbb{R}$, but, in $\mathbb{R}$, interval $[0, 1)$ is not an open set, interval $(0, 1]$ is not an open set, closed interval $[0, 1]$ is not an open set. The unit open circle $\{(x_1, x_2); x_1^2 + x_2^2 < 1\}$ is an open set in $\mathbb{R}^2$, but, in $\mathbb{R}^2$, the unit closed circle $\{(x_1, x_2); x_1^2 + x_2^2 \leqslant 1\}$ is not an open set.

The empty set $\varnothing$ and the whole space $\mathbb{R}$ are open.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$x \in E^\circ$ 指存在 $r > 0$ 使 $B(x, r) \subseteq E$。内部不是边界。若 $y \in B(x, r)$，取 $s = r - \|x - y\| > 0$，三角不等式给 $B(y, s) \subseteq B(x, r)$。因此球中每点都是内点，$E^\circ$ 开。内部的单调性来自球的包含，幂等性来自内部本身是开集。

45 原课件第 45 页：内点与内部



A correspondence $X \to X^\circ$ is called an interior operation if

(1) $X^\circ \subseteq X$,

(2) $X_1 \subseteq X_2 \Longrightarrow X_1^\circ \subseteq X_2^\circ$,

(3) $(X^\circ)^\circ = X^\circ$,

(4) $S^\circ = S$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“内点与内部”。完整定义、条件和论证见本讲原页第 44 页下方。

46 原课件第 46 页：内点与内部

Open Set

Theorem 1. If $E \subseteq \mathbb{R}^N$, then E° is an open set.

Proof. Let $E \subseteq \mathbb{R}^N$ and consider any $x_0 \in E^\circ$. Then $N_\delta(x_0) \subseteq E$ for some $\delta > 0$. We claim that $N_\delta(x_0) \subseteq E^\circ$, thus proving E° is open.

To establish the claim consider any $x \in N_\delta(x_0)$; we need to show that x is an interior point of E. Suppose $\rho(x, x_0) = r(< \delta)$. From the triangle inequality, $N_{\delta - r}(x) \subseteq N_\delta(x_0)$, and so $N_{\delta - r}(x) \subseteq E$, thus showing x is an interior point. Q.E.D.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“内点与内部”。完整定义、条件和论证见本讲原页第 44 页下方。

47 原课件第 47 页：有限开集的交

Open Set

Theorem 2. The intersection of finite open sets is an open set.

Proof. Define $G = \bigcap_{n=1}^{N} G_n$ where G_n is an open set for $n = 1, \cdots, N$. Note that G_n is an open set, $G_n^\circ = G_n$, then

$$G^\circ = \left(\bigcap_{n=1}^{N} G_n\right)^\circ = \bigcap_{n=1}^{N} G_n^\circ = \bigcap_{n=1}^{N} G_n = G.$$

Therefore G is an open set. Q.E.D.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

任取 $x \in \bigcap_{i=1}^{m} G_i$，各取 $r_i > 0$ 使 $B(x, r_i) \subseteq G_i$。有限多个正数的最小值 r 仍正，故 $B(x, r) \subseteq \bigcap_i G_i$。无限交不能照搬：$\bigcap_{n\geq 1}(-1/n, 1/n) = \{0\}$ 不开。

48 原课件第 48 页：有限开集的交

Open Set

Theorem 2. The intersection of finite open sets is an open set.

Proof. Define $G = \bigcap_{n=1}^{N} G_n$ where G_n is an open set for $n = 1, \cdots, N$. Suppose $x \in G$, then $x \in A_n$ for $n = 1, \cdots, N$, and for each n there exists $\delta_n > 0$ such that $N_{\delta_n}(x) \subseteq G_n$. Let $\delta = \min\{\delta_1, \cdots, \delta_N\}$. Then $\delta > 0$ and $N_\delta(x) \subseteq G$, implying G is open. Q.E.D.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“有限开集的交”。完整定义、条件和论证见本讲原页第 47 页下方。

49 原课件第 49 页：任意开集的并

Open Set

Theorem 3. The union of multiple open sets is an open set.

Proof. Define $G = \bigcup_{n=1}^{\infty} G_n$ where G_n is an open set for $n = 1, 2, \cdots$. Note $G_n \subseteq G$, then $G_n^\circ \subseteq G^\circ$. Thus $\bigcup_{n=1}^{\infty} G_n^\circ \subseteq G^\circ$. Note that G_n is an open set, $G_n^\circ = G_n$, then

$$G = \bigcup_{n=1}^{\infty} G_n = \bigcup_{n=1}^{\infty} G_n^\circ \subseteq G^\circ = \left(\bigcup_{n=1}^{\infty} G_n \right)^\circ .$$

Therefore G is an open set. Q.E.D.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

若 $x \in \bigcup_i G_i$，只需选一个含 x 的 G_j；由其开性取球 $B(x, r) \subseteq G_j \subseteq \bigcup_i G_i$。这不需要所有 G_i 共同的正半径，所以任意甚至不可数并仍开。与有限交比较，关键是“存在一个”与“对所有”的区别。

50 原课件第 50 页：任意开集的并

Open Set

Theorem 3. The union of multiple open sets is an open set.

Proof. Define $G = \bigcup_{n=1}^{\infty} G_n$ where G_n is an open set for $n = 1, 2, \cdots$. Suppose $x \in G$, then $x \in G_n$ for some n. For some such n choose $\delta_n > 0$ such that $N_{\delta_n}(x) \subseteq G_n$. Then certainly $N_{\delta_n}(x) \subseteq G$, and so G is open. Q.E.D.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“任意开集的并”。完整定义、条件和论证见本讲原页第 49 页下方。

51 原课件第 51 页：任意开集的并

Open Set

Theorem 3. If $\{G_\gamma : \gamma \in \Gamma\}$ is a family of open sets in $\mathbb{R}^N$, then its union $G = \bigcup_{\gamma\in\Gamma} G_\gamma$ is an open set.

Proof. Define $G = \bigcup_{\gamma\in\Gamma} G_\gamma$ where G_γ is an open set for $\gamma \in \Gamma$. Note $G_\gamma \subseteq G$, then $G_\gamma^\circ \subseteq G^\circ$. Thus $\bigcup_{\gamma\in\Gamma} G_\gamma^\circ \subseteq G^\circ$. Note that G_γ is an open set, $G_\gamma^\circ = G_\gamma$, then

$$G = \bigcup_{\gamma\in\Gamma} G_\gamma = \bigcup_{\gamma\in\Gamma} G_\gamma^\circ \subseteq G^\circ = \left(\bigcup_{\gamma\in\Gamma} G_\gamma\right)^\circ .$$

Therefore G is an open set. Q.E.D.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“任意开集的并”。完整定义、条件和论证见本讲原页第 49 页下方。

52 原课件第 52 页：任意开集的并

Open Set

Theorem 3. If $\{G_\gamma : \gamma \in \Gamma\}$ is a family of open sets in $\mathbb{R}^N$, then its union $G = \bigcup_{\gamma\in\Gamma} G_\gamma$ is an open set.

Proof. Define $G = \bigcup_{\gamma\in\Gamma} G_\gamma$ where G_γ is an open set for $\gamma \in \Gamma$.
Suppose $x \in G$, then $x \in G_\gamma$ for some $\gamma \in \Gamma$. For some such γ choose $\delta_\gamma > 0$ such that $N_{\delta_\gamma}(x) \subseteq G_\gamma$. Then certainly $N_{\delta_\gamma}(x) \subseteq G$, and so G is open. Q.E.D.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“任意开集的并”。完整定义、条件和论证见本讲原页第 49 页下方。

53 原课件第 53 页：实开集的区间分解



The following important theorems reveal the structure of open sets and give us a further perceptual understanding of open sets.

Theorem 4

(1) The non-empty open set in $\mathbb{R}$ is the union of countable disjoint open intervals (here also includes $(-\infty, a)$, (b, ∞) and $(-\infty, \infty)$);

(2) The non-empty open set G in $\mathbb{R}^N$ is the union of countable disjoint semi-open and semi-closed rectangulars.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

对 $a \in G$，把包含 a 且包含于 G 的开区间取并，得极大开区间 I_a。两个这样的区间相交，则并仍是区间，因此极大性使其相等。故不同 I_a 不交且覆盖 G。每个非空区间含有理数，选其中一个；不同区间对应不同有理数，故只有可数多个区间。端点允许无穷。

54 原课件第 54 页：实开集的区间分解

Open Set

Proof of (1). Let G be an open set in $\mathbb{R}$. For any point $a \in G$, since a is the interior point of G, there is $\delta > 0$ such that $(a-\delta, a+\delta) \subseteq G$. Now define

$$a' = \inf\{x : (x, a) \subseteq G\} \qquad \text{and} \qquad a'' = \sup\{x : (a, x) \subseteq G\},$$

(here, a' can be $-\infty$, and a'' can be $+\infty$). obviously, $a' < a < a''$ and $(a', a'') \subseteq G$. In fact, for any point z in the interval (a', a''), we assume $a' < z \leqslant a$, there must be x such that $a' < x < z$ and $(x, a) \subseteq G$, that is, $z \in G$. We call such an open interval (a', a'') the constituent interval I_a of G (about point a).

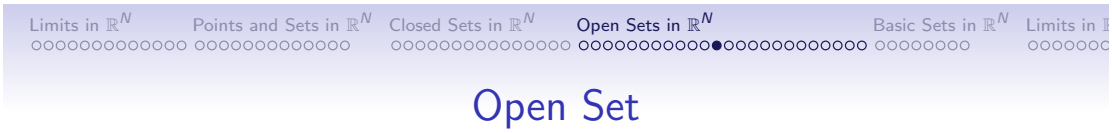
If $I_{a_1} = (a_1', a_1'')$, $I_{a_2} = (a_2', a_2'')$ is the constituent interval of G, then they are coincident or disjoint. For this purpose, we assume $a_1 < a_2$. If $I_{a_1} \cap I_{a_2} \neq \varnothing$, then there is $a_2' < a_1''$, then define $\min\{a_1', a_2'\} = a_0$ and $\max\{a_1'', a_2''\} = a^0$, then $(a_0, a^0) = (a_1', a_1'') \cup (a_2', a_2'')$. Take $x \in I_{a_1} \cap I_{a_2}$, then $I_x = (a_0, a^0)$ is a constituent interval and $(a_0, a^0) = (a_1', a_1'') = (a_2', a_2'')$.

Finally, we know that the family of disjoint intervals in $\mathbb{R}$ are countable.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“实开集的区间分解”。完整定义、条件和论证见本讲原页第 53 页下方。

55 原课件第 55 页：欧氏开集的方块分解



Proof of (2). Firstly, $\mathbb{R}^N$ is divided into countable semi-open and semi-closed cubes with side length of 1 by lattice points (coordinates are all integers), and the whole set is denoted as Γ_0. Each side of each cube in Γ_0 is bisected, then each cube can be divided into 2^N semi-open and semi-closed cubes with a side length of $\frac{1}{2}$, In Γ_0, all the sub-cubes thus created are denoted Γ_1. Continue to divide by this method to obtain the sequence $\{\Gamma_k; k \in \mathbb{N}\}$, where the side length of each cube in Γ_k is 2^{-k}, and this cube is the union of the corresponding 2^k disjoint cubes in Γ_{k+1}. We call the cube thus divided a dyadic cube.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

把空间用边长 2^{-k} 的半开方块分割，保留完全包含于 G 且未被更粗已选方块覆盖的方块。每层可数，总层数可数；嵌套网格保证所选方块两两不交。给定 $x \in G$，存在球包含于 G，足够细的含 x 方块直径小于该球半径，所以 x 最终被选中方块覆盖。半开约定避免公共边界重复。

56 原课件第 56 页：欧氏开集的方块分解

Open Set

Now put all cubes in Γ_0 included in G are taken out and written as $\mathscr{H}_0$. Then put all cubes in Γ_1 included in $G \setminus \bigcup_{H\in\mathscr{H}_0} H$ are taken out and written as $\mathscr{H}_1$ ($H \in \mathscr{H}_0$ is semi-open and semi-closed cube). And so on, $\mathscr{H}_K$ is the whole set of semi-open and semi-closed cubes in Γ_K included in $G \setminus \bigcup_{k=1}^{K-1} \bigcup_{H\in\mathscr{H}_k} H$. The whole set of semi-open and semi-closed cubes in $\mathscr{H}_K$ is countable for $K = 0, 1, 2, \cdots$. Since G is an open set, for any $x \in G$, there is $\delta > 0$ such that $B(x, \delta) \subseteq G$. The diameter of the cube in Γ_K approaches to zero as $K \to \infty$, so x will eventually fall into a certain cube in Γ_K. Thus

$$G = \bigcup_{k=0}^{\infty} \bigcup_{H\in\mathscr{H}_k} H, \quad H \text{ denotes semi-open and semi-closed dyadic cubes.}$$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“欧氏开集的方块分解”。完整定义、条件和论证见本讲原页第 55 页下方。

57 原课件第 57 页：可数拓扑基

Open Set

There exists another important fact about the open sets in $\mathbb{R}^N$, that is, there is an open set family Γ, which consists of countable open sets in $\mathbb{R}^N$ such that any open set in $\mathbb{R}^N$ is the union of some open sets in Γ. In fact, Γ can be taken as

$$\left\{ B\left(x, \frac{1}{K}\right) \mid x \text{ is a rational point in } \mathbb{R}^N, \text{ and } K \text{ is a natural number} \right\}$$

Firstly, Γ is a countable set. Secondly, for any point x in the open set G in $\mathbb{R}^N$, there must be $\delta > 0$ such that $B(x, \delta) \subseteq G$. Now take the rational point x' such that $d(x, x') < \frac{1}{K}$, where $K > \frac{2}{\delta}$ such that

$$x \in B\left(x', \frac{1}{K}\right) \subseteq B(x, \delta) \subseteq G.$$

Obviously, the union of such $B\left(x', \frac{1}{K}\right)$ is G. Q.E.D.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

所有有理中心、正有理半径的球构成可数族。若 $x \in G$ 开，选 $B(x, r) \subseteq G$，再选有理中心 q 足够近于 x，及有理半径 s 满足 $\|x - q\| < s < r - \|x - q\|$，便有 $x \in B(q, s) \subseteq G$。因此任何开集都是某些基球的并。

58 原课件第 58 页：开集习题与必要纠错

Open Set

Questions. Answer the following questions:

1. If $N \geqslant 2$, $F \subset \mathbb{R}^N$ ($F \neq \mathbb{R}^N$) is a closed set, then ∂F is an infinite set.
2. Let $E \subseteq \mathbb{R}^N$, prove that $E^\circ = \mathbf{C}\left(\overline{\mathbf{C}E}\right)$ and $\partial E = \overline{E} \setminus E^\circ$.
3. Prove that the set of discontinuous points of function $f(x_1, x_2) = \begin{cases} x_1 \sin \frac{1}{x_2}, & x_2 \neq 0, \\ 0, & x_2 = 0 \end{cases}$ is not a closed set.
4. Prove that $G \subseteq \mathbb{R}^N$ is an open set if and only if $G \cap \partial G = \varnothing$; $F \subseteq \mathbb{R}^N$ is a closed set if and only if $\partial F \subseteq F$.
5. Let $G \subseteq \mathbb{R}^N$ be a non-empty open set, $r_0 > 0$. If for any $x \in G$, define a closed sphere $\overline{B}(x, r_0)$, Prove that $A = \bigcup_{x \in G} \overline{B}(x, r_0)$ is an open set.
6. Let $F \subseteq \mathbb{R}$ be a closed set, prove that there is a countable subset of E in F such that $\overline{E} = F$.
7. Let every point in $E \subseteq \mathbb{R}^N$ be an isolated point of E, prove that E is the intersection of an open set and a closed set.
8. Let $f \in C([a, b])$, and $f'(x) > 0$ except $\{x_n\} \subseteq [a, b]$, prove that $f(x)$ is strictly increasing.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

1. 原命题假：$F = \{0\} \subset \mathbb{R}^N$ 闭而边界只有一个点。2. x 不在补集闭包等价于存在球避开补集，即 $x \in E^\circ$；由此 $\partial E = \overline{E} \setminus E^\circ$。3. 当 $x_2 \neq 0$ 函数连续；$(a, 0)$ 且 $a \neq 0$ 时沿正弦取 ± 1 的序列得不同极限，在原点则 $|f| \leq |x_1| \to 0$，故不连续点为 $(\mathbb{R} \setminus \{0\}) \times \{0\}$，不闭。4. 直接用内部、边界、闭包的分解。5. 若 $y = x + v$ 且 $\|v\| \leq r_0$，由 G 开，$x + h \in G$ 对足够小 h 成立，因而 $y + h = (x + h) + v \in A$，故 A 开。6. 对每个与 F 相交的基球选一个 F 中点，所得集可数且在 F 中稠密；F 闭保证闭包等于 F。7. 离散集 E 不交其导集 E'，且 E' 闭，故 $E = \overline{E} \cap (\mathbb{R}^N \setminus E')$。8. 先证不减：若存在 $a < b$ 使 $f(a) > f(b)$，对每个 $y \in (f(b), f(a))$，令 t_y 是 $[a, b]$ 中满足 $f(t_y) = y$ 的最后一点。连续性及介值定理保证它存在且在 (a, b)，并且在其右侧一直有 $f(t) < y$，否则连续性又给更晚的同水平点。若 t_y 处可导，则右差商非正，故 $f'(t_y) \leq 0$，所以 t_y 必在可数例外集。不同 y 给不同 t_y，这把不可数实区间单射到可数集，矛盾。故 f 不减。若 $f(a) = f(b)$，不减性使整个 $[a, b]$ 上恒定；从区间选一个不在可数例外集中的内点，该点导数为0，与假设正导数矛盾。因此严格递增。本题不必要求例外点也可导，但必须保持全区间连续。

59 原课件第 59 页：紧性与有限子覆盖

Open Set

Let $E \subseteq \mathbb{R}^N$, Γ is an family of open sets in $\mathbb{R}^N$. If for any $x \in E$, there exists $G \in \Gamma$ such that $x \in G$, then Γ is called an open covering of E. Let Γ be an open covering of E. If $\Gamma' \subseteq \Gamma$ is still an open covering of E, then Γ' is a subcovering of Γ (about E).

Lemma Any open cover Γ of the set $E \subseteq \mathbb{R}^N$ contains a countable subcovering.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

开覆盖是开集族，其并包含目标集合；有限子覆盖必须从原覆盖取有限成员。有限维中闭有界等价于紧。证明闭有界给有限子覆盖：若没有，把含 E 的闭方块不断二分，每次保留不能有限覆盖的一个子方块。得到直径趋零的套方块及 $x_n \in E$，其极限 $x \in E$（用闭性）。某原覆盖开集含 x，故含 $B(x,r)$；足够小的整个所选方块都落入该球，可由一个开集覆盖，矛盾。另一证明的 Lebesgue 数步骤：若无统一半径，取 x_n 使 $B(x_n, 1/n)$ 不被单个覆盖成员包含；抽子列 $x_n \to x$，取 $B(x,r) \subset U$，当 $\|x_n - x\| < r/2, 1/n < r/2$ 时矛盾。闭性和有界性缺一不可：$(0,1)$ 可被 $(1/n, 1)$ 覆盖却不能有限覆盖，$\mathbb{N}$ 的局部小区间覆盖也不能有限覆盖。

60 原课件第 60 页：紧性与有限子覆盖

Open Set

Theorem 5 (Heine-Borel Finite Covering Theorem) An open covering of a bounded closed set in $\mathbb{R}^N$ contains a finite subcovering.

Proof Let F is a bounded closed set in $\mathbb{R}^N$, and Γ is an open covering of F. From the above Lemma, it can be assumed that Γ consists of countable open sets:

$$\Gamma = \{G_1, \cdots, G_K, \cdots\}.$$

Define

$$H_K = \bigcup_{k=1}^{K} G_k, \qquad L_K = F \cap \mathbf{C}H_K \quad k = 1, 2, \cdots.$$

Obviously, H_K is an open set, L_K is a closed set with $L_K \supseteq L_{K+1}$ for $k = 1, 2, \cdots$. There are two cases:

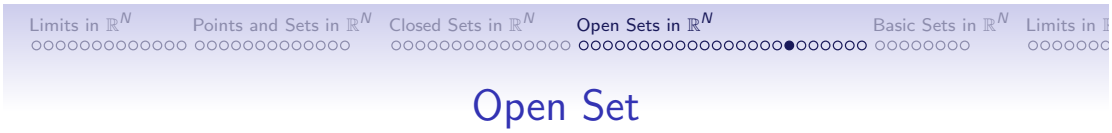
(1) there is K_0 such that L_{K_0} is an empty set, that is, $\mathbf{C}H_{K_0}$ does not contain any point of F, then $F \subseteq H_{K_0}$, the theorem is proved;

(2) If all L_K is not an empty set, then, from Cantor's closed set theorem, there is a point of $x_0 \in L_K$ for $k = 1, 2, \cdots$, i.e. $x_0 \in F$ and $x_0 \in \mathbf{C}H_K$ for $k = 1, 2, \cdots$. Thus there is a point $x_0 \in F$ does not belong to everything H_K, which is inconsistent with the original assumption, so case (2) does not exist. Q.E.D.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“紧性与有限子覆盖”。完整定义、条件和论证见本讲原页第 59 页下方。

61 原课件第 61 页：紧性与有限子覆盖



Theorem 5 (Heine-Borel Finite Covering Theorem) If $\mathscr{H}$ is an open covering of a closed and bounded subset E of $\mathbb{R}^N$ (for any x in E there exists a neighborhood $H \in \mathscr{H}$ such that $x \in H$); then E has an open covering $\tilde{\mathscr{H}}$ consisting of finitely many open sets belonging to $\mathscr{H}$.

Proof. The proof method belongs to Lebesgue. We complete the proof by two steps.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“紧性与有限子覆盖”。完整定义、条件和论证见本讲原页第 59 页下方。

62 原课件第 62 页：紧性与有限子覆盖

Open Set

(1) There exists $\delta > 0$ such that any δ neighborhood $N(x, \delta)$ of center of x in E is contained in some open neighborhood in $\mathscr{H}$. We use reduction to absurdity to check this setting, and assume that there isn't such $\delta > 0$. For any natural number n, we cannot take $\frac{1}{n}$ as δ, then there exists $x_n \in E$ such that $N\left(x_n, \frac{1}{n}\right)$ does not contain in any open neighborhood in $\mathscr{H}$. E is bounded, $\{x_n : n = 1, 2, \cdots\} \subseteq E$ is also bounded. If $\{x_n : n = 1, 2, \cdots\}$ has infinite different members, then there exists an accumulation points x_0 from Bolzano - Weierstrass Theorem. If $\{x_n : n = 1, 2, \cdots\}$ has only finite different members, then there exists at least a point x_0 presented infinite times. In a word, $\{x_n : n = 1, 2, \cdots\} \subseteq E$ has a convergent subsequence $\{x_{n_k} : n = 1, 2, \cdots\} \subseteq E$ with $\lim_{k\to\infty} x_{n_k} = x_0$. Note that E is a closed set, then $x_0 \in E$. Since $\mathscr{H}$ is an open covering of a closed and bounded set E, then there exists $N \in \mathscr{H}$ such that $x_0 \in N$. Define $N = N(y_0, \eta)$, then there exists $\eta' = \min\{\eta, \rho(x_0, y_0)\} > 0$ such that $N(x_0, \eta') \subseteq N(y_0, \eta)$. Note $\lim_{k\to\infty} x_{n_k} = x_0$, then we can take enough large natural number n_k such that $\rho(x_{n_k}, x_0) < \frac{\eta'}{2}$ and $\frac{1}{n_k} < \frac{\eta'}{2}$, then $N\left(x_{n_k}, \frac{1}{n_k}\right) \subseteq N(x_0, \eta') \subseteq N(y_0, \eta) \in \mathscr{H}$. This contradicts the definition of x_{n_k}. Therefore there exists such $\delta > 0$, which is called Lebesgue number.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“紧性与有限子覆盖”。完整定义、条件和论证见本讲原页第 59 页下方。

63 原课件第 63 页：紧性与有限子覆盖

Open Set

(2) Since E is bounded, we can divide E into finite small cubes by using hyperplanes parallel axises such that the length in the each cube is strictly less than $\delta > 0$. Set these small cubes are $E_1, \cdots, E_K$ with $E = \sum_{k=1}^{K} E_k$, we take arbitrary $x_k \in F_k$ such that $F_k \subseteq N(x_k, \delta)$, then there exists $N_k \in \mathscr{H}$ such that $N(x_k, \delta) \subseteq N_k$. Thus we obtain finite open neighborhoods $N_1, \cdots, N_K$ with $E = \sum_{k=1}^{K} E_k \subseteq \bigcup_{k=1}^{K} N(x_k, \delta) \subseteq \bigcup_{k=1}^{K} N_k$. Q.E.D.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“紧性与有限子覆盖”。完整定义、条件和论证见本讲原页第 59 页下方。

64 原课件第 64 页：紧性与有限子覆盖

Open Set

Note that in the above theorem, the boundedness condition cannot be absent. For example, in $\mathbb{R}$, there is no finite subcovering for open covering $\left\{\left(n-\frac{1}{2}, n+\frac{1}{2}\right)\right\}$ of natural number set.
Similarly, the conditions of a closed set are also indispensable. For example, in $\mathbb{R}$, for the point set $\left\{1, \frac{1}{2}, \cdots, \frac{1}{n}, \cdots\right\}$, define an open covering

$$\left\{\left(\frac{1}{n}-\frac{1}{2n}, \frac{1}{n}+\frac{1}{2n}\right)=\left(\frac{1}{2n}, \frac{3}{2n}\right); n \in \mathbb{N}\right\},$$

there is no finite subcovering.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“紧性与有限子覆盖”。完整定义、条件和论证见本讲原页第 59 页下方。

65 原课件第 65 页：紧集的邻域与逆定理

Open Set

Example: If F is a bounded closed set in $\mathbb{R}^N$, and G is an open set in $\mathbb{R}^N$ and $F \subseteq G$, then there is $\delta > 0$ such that when $|x| < \delta$,

$$F + \{x\} \equiv \{y + x \mid y \in F\} \subseteq G.$$

Proof. For any $y \in F$, as $y \in G$, there is $\delta_y > 0$ such that $B(y, \delta_y) \subseteq G$. As $\left\{ B\left(y, \frac{\delta_y}{2}\right) \mid y \in F \right\}$ constitutes an open covering of F, according to the finite sub-covering theorem, there is $y_1, \cdots, y_K \in F$ such that

$$F \subseteq \bigcup_{k=1}^{K} B\left(y_k, \frac{\delta_{y_k}}{2}\right).$$

Thus, each $y \in F$ belongs to at least one $B\left(y_k, \frac{\delta_{y_k}}{2}\right)$, and the distance between y and any point z in $\mathbf{C}G$ is

$$|y - z| \geqslant |z - y_k| - |y_k - y| > \delta_{y_k} - \frac{\delta_{y_k}}{2} = \frac{\delta_{y_k}}{2}.$$

Take $\delta = \frac{1}{2} \min\{\delta_{y_1}, \cdots, \delta_{y_K}\}$, then, when $|x| < \delta$, $y + x \in G$, that is, $F + \{x\} \subseteq G$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

若紧 $F \subset G$ 开，给每个 $y \in F$ 选 $B(y, r_y) \subset G$，用半径 $r_y/2$ 的球有限覆盖 F，取有限 $r_y/2$ 的最小值 $\delta > 0$。任意 $\|h\| < \delta$ 都使 $F + h \subset G$。反之紧集必有界，因为球 $B(0, n)$ 覆盖它且有限子覆盖给统一半径；必闭，因为对 $z \notin E$，以 $x \in E$ 为中心取半径 $\|x - z\|/3$ 的球，有限覆盖后可在 z 周围找到避开 E 的正半径球。

66 原课件第 66 页：紧集的邻域与逆定理

Open Set

Theorem 6. Let $E \subseteq \mathbb{R}^N$. If any open covering of E contains finite sub-coverings, then E is a bounded closed set.

Proof. If $y \in \mathbf{C}E$, then for every $x \in E$, there is $\delta_x > 0$ such that

$$B(x, \delta_x) \cap B(y, \delta_x) = \varnothing.$$

Obviously, $\{B(x, \delta_x) : x \in E\}$ is an open covering of E. Then there are a finite sub-covering, written as

$$B(x_1, \delta_{x_1}), \cdots, B(x_K, \delta_{x_K}),$$

then E is a bounded set. Now define $\delta_0 = \min\{\delta_{x_1}, \cdots, \delta_{x_K}\}$, then $B(y, \delta_0) \cap E = \varnothing$, that is, $y \notin E'$. Thus $E' \subseteq E$. E is a closed set and hence boundedness is obvious. Q.E.D.

Note that if any open covering of E contains a finite sub-covering, we call E a compact set. The above two theorems show that a compact set in $\mathbb{R}^N$ is a bounded closed set.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“紧集的邻域与逆定理”。完整定义、条件和论证见本讲原页第 65 页下方。

67 原课件第 67 页：紧性习题

Open Set

Questions. Answer the following questions:

1. In $\mathbb{R}^N$, $\{G_\gamma; \gamma \in \Gamma\}$ is an open covering of E. Can $\{\overline{G_\gamma}; \gamma \in \Gamma\}$ cover $\overline{E}$?
2. Let $\Gamma = \{[a_\alpha, b_\alpha] : \alpha \in [0,1]\}$, and any two closed intervals in Γ must intersect. Prove that
$$\bigcap_{\alpha\in[0,1]} [a_\alpha, b_\alpha] \neq \varnothing.$$
3. Let $F \subseteq \mathbb{R}$ be a non-empty countable closed set, prove that F must contain isolated points.
4. Let $\{f_n(x)\}$ be a sequence of non-negative decreasing continuous functions on $\mathbb{R}$. If $f_n(x) \to 0$ as $n \to \infty$ on a bounded closed set F, prove $f_n(x)$ uniformly converges to zero on F.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

1. 原覆盖的闭包族当然仍覆盖 E，因为每个成员都扩大。2. 任意两闭区间相交给出 $a_\alpha \le b_\beta$ 对所有指标成立；故 $s = \sup_\alpha a_\alpha \le b_\beta$，又 $a_\beta \le s$，所以 s 属于全部区间。任取固定区间可保证确界有限。3. 若非空闭集无孤立点，在每个已有小闭区间内可选两个不同的集内点及互不交更小闭区间；递归构造二叉树并令直径趋零。每条无限0–1路径由完备性给一个集内极限，不同路径给不同点，从而不可数，矛盾。4. 对 $\varepsilon > 0$，开集 $U_n = \{x : f_n(x) < \varepsilon\}$ 递增并覆盖紧 F，有限子覆盖可合为某个 U_N。故所有 $n \ge N, x \in F$ 有 $0 \le f_n(x) < \varepsilon$，这正是一致收敛（Dini 型证明）。

68 原课件第 68 页：闭包与内部的抽象公理

Close set and Open Set

If with each subset X of a set S we associate a subset $\overline{X}$ of S such that

(1) $X \subseteq \overline{X}$,

(2) $X_1 \subseteq X_2 \Longrightarrow \overline{X}_1 \subseteq \overline{X}_2$,

(3) $\overline{\overline{X}} = \overline{X}$,

(4) $\overline{\varnothing} = \varnothing$,

then we call the correspondence $X \to \overline{X}$ a closure operation.

In a similar manner, if with each subset X of a set S we associate a subset X° of S such that

(1) $X^\circ \subseteq X$,

(2) $X_1 \subseteq X_2 \Longrightarrow X_1^\circ \subseteq X_2^\circ$,

(3) $(X^\circ)^\circ = X^\circ$,

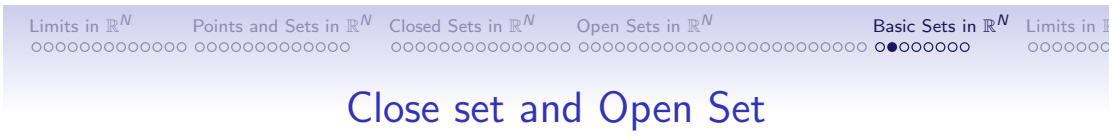
(4) $S^\circ = S$.

then we call the correspondence $X \to X^\circ$ a interior operation.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

闭包算子 c 满足扩张、单调、幂等及 $c(\varnothing)=\varnothing$；定义 $i(A)=S\setminus c(S\setminus A)$。补集反向包含把扩张变为收缩，单调仍为单调，幂等由两次取补消去得到，$i(S)=S$。反向公式 $c(A)=S\setminus i(S\setminus A)$，因此两构造互逆。闭集是闭包不动集：任意闭集交 F 满足 $c(F)\subseteq c(F_j)=F_j$ 对所有 j 成立，加扩张性得相等。若 $A\subset F$ 闭，则 $c(A)\subset F$；且 $c(A)$ 自己闭，所以它是所有含 A 闭集的交。注意这四条是较一般闭包系统公理，未必单独推出有限闭并性质，欧氏拓扑还有相应额外性质。

69 原课件第 69 页：闭包与内部的抽象公理



The sets F such that $F = \overline{F}$ are said to be closed and the sets G such that $G = G^\circ$ are said to be open, with respect to the appropriate operations in each case.

Example. If S denotes the real line $\mathbb{R}$ and $\overline{X}$ the smallest segment $[a, b]$ which contains all the points of the set X, then the correspondence $X \to \overline{X}$ is a closure operation.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“闭包与内部的抽象公理”。完整定义、条件和论证见本讲原页第 68 页下方。

70 原课件第 70 页：闭包与内部的抽象公理

Close set and Open Set

Theorem 1. A one-one correspondence can be established between closure operations and interior operations by means of the formulae:

$$X^\circ = \mathbf{C}(\overline{\mathbf{C}X}) \qquad \text{and} \qquad \overline{X} = \mathbf{C}(\mathbf{C}X)^\circ$$

Proof. The correspondence $X \to X^\circ = \mathbf{C}(\overline{\mathbf{C}X})$ is clearly an interior operation, for

(1) $X^\circ = \mathbf{C}(\overline{\mathbf{C}X}) \subseteq \mathbf{C}(\mathbf{C}X) = X$,

(2) $X_1 \subseteq X_2 \Longrightarrow \mathbf{C}X_1 \supseteq \mathbf{C}X_2 \Longrightarrow (\overline{\mathbf{C}X_1}) \supseteq (\overline{\mathbf{C}X_2}) \Longrightarrow X_1^\circ \subseteq X_2^\circ$,

(3) $(X^\circ)^\circ = \mathbf{C}(\overline{\overline{\mathbf{C}X}}) = \mathbf{C}(\overline{\mathbf{C}X}) = X^\circ$,

(4) $S^\circ = \mathbf{C}\overline{\varnothing} = \mathbf{C}\varnothing = S$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“闭包与内部的抽象公理”。完整定义、条件和论证见本讲原页第 68 页下方。

71 原课件第 71 页：闭包与内部的抽象公理

Close set and Open Set

The correspondence $X \to \overline{X} = \mathbf{C}(\mathbf{C}X)^\circ$ is clearly a closure operation, for

(1) $\overline{X} = \mathbf{C}(\mathbf{C}X)^\circ \supseteq \mathbf{C}(\mathbf{C}X) = X$,

(2) $X_1 \subseteq X_2 \Longrightarrow \mathbf{C}X_1 \supseteq \mathbf{C}X_2 \Longrightarrow (\mathbf{C}X_1)^\circ \supseteq (\mathbf{C}X_2)^\circ \Longrightarrow \overline{X}_1 \subseteq \overline{X}_2$,

(3) $\overline{\overline{X}} = \mathbf{C}[(\mathbf{C}X)^\circ]^\circ = \mathbf{C}(\mathbf{C}X)^\circ = \overline{X}$,

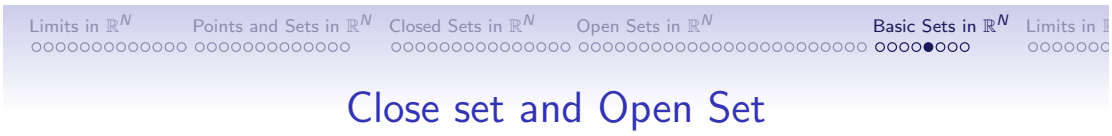
(4) $\overline{\varnothing} = \mathbf{C}S^\circ = \mathbf{C}S = \varnothing$.

It is easily shown that the correspondence thus set up between the two kinds of operations is one-one. Q.E.D.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“闭包与内部的抽象公理”。完整定义、条件和论证见本讲原页第 68 页下方。

72 原课件第 72 页：闭包与内部的抽象公理



Remark. The complement of an open set is a closed set (and vice versa), for $\mathbf{C}F = \mathbf{C}\overline{F} = (\mathbf{C}F)^\circ$; in what follows, we shall study only properties of closure operations, those of interior operations being deducible immediately on appealing to duality, for $\mathbf{C}G = \mathbf{C}G^\circ = \overline{\mathbf{C}G}$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“闭包与内部的抽象公理”。完整定义、条件和论证见本讲原页第 68 页下方。

73 原课件第 73 页：闭包与内部的抽象公理

Close set and Open Set

Theorem 2. The intersection of a family $\mathscr{F} = (F_\gamma \mid \gamma \in \Gamma)$ of closed sets is a closed set.

Proof. Writing $F = \bigcap_{\gamma\in\Gamma} F_\gamma$, we have, by condition (2) for a closure operation,

$$\overline{F} \subseteq \overline{F}_\gamma = F_\gamma$$

for each γ. Therefore $\overline{F} \subseteq F$; hence, by condition (1), $\overline{F} = F$. Q.E.D.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“闭包与内部的抽象公理”。完整定义、条件和论证见本讲原页第 68 页下方。

74 原课件第 74 页：闭包与内部的抽象公理

Close set and Open Set

Theorem 2. The union of a family $\mathscr{G} = (G_\gamma \mid \gamma \in \Gamma)$ of open sets is an open set.

Proof. Writing $G = \bigcup_{\gamma\in\Gamma} G_\gamma$, we have, by condition (2) for a interior operation,

$$G_\gamma = G_\gamma^\circ \subseteq G^\circ$$

for each γ. Therefore $G \subseteq G^\circ$; hence, by condition (1), $G^\circ = G$. Q.E.D.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“闭包与内部的抽象公理”。完整定义、条件和论证见本讲原页第 68 页下方。

75 原课件第 75 页：闭包与内部的抽象公理

Close set and Open Set

Theorem 3. The closure $\overline{X}$ of X is the intersection of all the closed sets which contain X.

Proof. Let E be the intersection of all the closed sets which contain X; by the preceding theorem, E is closed. Since $\overline{X}$ is a closed set which contains X, we have $E \subseteq \overline{X}$. But we also have $E = \overline{E} \supseteq \overline{X}$, by condition (2), whence $E = \overline{X}$. Q.E.D.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“闭包与内部的抽象公理”。完整定义、条件和论证见本讲原页第 68 页下方。

76 原课件第 76 页：闭包的序列判据与相对闭性

Limits in $\mathbb{R}^N$

Let X be a subset of $\mathbb{R}^N$; a point $x \in \mathbb{R}^N$ is adherent to X if there is a sequence of points of X tending to x. One can also say, in a looser manner, that x is adherent to X if there are points of X arbitrarily close to x. Any point x of X is adherent to X; it suffices to take the sequence whose points are all equal to x.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

$x \in \overline{A}$ 当且仅对任意 n 可选 $a_n \in A$ 使 $\|x - a_n\| < 1/n$。闭集即任何集内收敛序列的极限仍在集内。闭包幂等可用二段逼近：先在 $\overline{A}$ 找距 x 小于 $\varepsilon/2$ 的点，再在 A 找距它小于 $\varepsilon/2$ 的点。任意闭交用序列同时属于每个因子；有限闭并则用有限抽屉原理找到无限多项属于某一个闭集的子列。相对闭包为 $S \cap \overline{A}$，只保留属于背景空间 S 的极限；例如 $(0,1)$ 在自身中闭，却不在 $\mathbb{R}$ 中闭。

77 原课件第 77 页：闭包的序列判据与相对闭性

Limits in $\mathbb{R}^N$

The set of the points of $\mathbb{R}^N$ adherent to X is called the adherence (often also the closure) of X, and denoted $\overline{X}$. According to the last remark of 1.6.g:

(1) $X \subseteq \overline{X}$.

It is also clear that:

(2) $X \subseteq X'$ implies $\overline{X} \subseteq \overline{X'}$.

Example: (1) Let M be a positive real number, and consider, in $\mathbb{R}^2$, the square (with perimeter excluded) $X = \{x \in \mathbb{R}^2 \mid |x_n| < M \text{ for } n = 1, 2\}$; then $\overline{X} = \{x \in \mathbb{R}^2 \mid |x_n| \leqslant M \text{ for } n = 1, 2\}$. (2) Using (2) of 1.5.l, one obtains $\overline{\mathbb{Q}} = \mathbb{R}$: the adherence of the set of rationals is the set of reals.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“闭包的序列判据与相对闭性”。完整定义、条件和论证见本讲原页第 76 页下方。

78 原课件第 78 页：闭包的序列判据与相对闭性

Limits in $\mathbb{R}^N$

A subset X of $\mathbb{R}^N$ is closed if it owns its adherent points. This may be expressed as $\overline{X} \subseteq X$. Thus, because of (1), X is closed if and only if it is equal to its adherence, $X = \overline{X}$. One may also say that a set is closed if and only if every point at zero distance from the set belongs to the set; this requirement will often be imposed below.

Example: (1) In the examples of 1.6.h. X, the subset of $\mathbb{R}^2$, and $\mathbb{Q}$, the subset of $\mathbb{R}$, are not dosed; $\overline{X}$ is closed. (2) The subset of $\mathbb{R}$, $\left\{0, 1, \frac{1}{2}, \frac{1}{3}, \cdots, \frac{1}{q}, \cdots\right\}$, is closed.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“闭包的序列判据与相对闭性”。完整定义、条件和论证见本讲原页第 76 页下方。

79 原课件第 79 页：闭包的序列判据与相对闭性

Limits in $\mathbb{R}^N$

One can prove:

(3) If X is a subset of $\mathbb{R}^N$, its adherence $\overline{X}$ is closed.

Proof: For any $\varepsilon > 0$ and any $x \in \overline{\overline{X}}$, there exists $x_\varepsilon \in \overline{X}$ such that $d(x, x_\varepsilon) < \frac{\varepsilon}{2}$ and then there exists $x'_\varepsilon \in X$ such that $d(x_\varepsilon, x'_\varepsilon) < \frac{\varepsilon}{2}$, hence $d(x, x'_\varepsilon) \leqslant d(x, x_\varepsilon) + d(x_\varepsilon, x'_\varepsilon) < \varepsilon$, which implies $x \in \overline{X}$. Hence $\overline{\overline{X}} \subseteq \overline{X}$. $\overline{X}$ is closed. □

That is, $\overline{X} = \overline{\overline{X}}$. Since every closed set containing X clearly contains $\overline{X}$, the adherence of X can be characterized as the smallest closed set containing X.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“闭包的序列判据与相对闭性”。完整定义、条件和论证见本讲原页第 76 页下方。

80 原课件第 80 页：闭包的序列判据与相对闭性

Limits in $\mathbb{R}^N$

It is easy to prove that:

(4) If $\mathscr{X}$ is a set of closed subsets of $\mathbb{R}^N$, their intersection, $\bigcap_{X\in\mathscr{X}} X$, is a closed subset of $\mathbb{R}^N$.

Proof: For any $x \in \overline{\bigcap_{X\in\mathscr{X}} X}$, there exists $\{x^q; q \in \mathbb{N}\} \subseteq \bigcap_{X\in\mathscr{X}} X$ such that $x^q \to x$. And for any $X \in \mathscr{X}$, $\{x^q; q \in \mathbb{N}\} \subseteq X$, which implies $x \in \overline{X}$ for any $X \in \mathscr{X}$. As X is closed, $x \in X$ for any $X \in \mathscr{X}$. Then $x \in \bigcap_{X\in\mathscr{X}} X$, which implies $\overline{\bigcap_{X\in\mathscr{X}} X} \subseteq \bigcap_{X\in\mathscr{X}} X$. So $\bigcap_{X\in\mathscr{X}} X$ is a closed subset of $\mathbb{R}^N$. $\square$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“闭包的序列判据与相对闭性”。完整定义、条件和论证见本讲原页第 76 页下方。

81 原课件第 81 页：闭包的序列判据与相对闭性

Limits in $\mathbb{R}^N$

It is a little more difficult to prove that:

(5) If $X_1, \cdots, X_n, \cdots, X_N$ ore N closed subsets of $\mathbb{R}^N$, their union, $\bigcup_{n=1}^{N} X_n$, is a closed subset of $\mathbb{R}^N$.

Proof: For any $x \in \overline{\bigcup_{n=1}^{N} X_n}$, there exists $\{x^q; q \in \mathbb{N}\} \subseteq \bigcup_{n=1}^{N} X_n$ such that $x^q \to x$. So there exists $n_0 \in \{1, \cdots, N\}$ such that there are infinite elements $\{x^{q_k}; q_k \in \mathbb{N}\}$ of $\{x^q; q \in \mathbb{N}\}$ in X_{n_0}, hence $x^{q_k} \to x$, which implies $x \in \overline{X}_{n_0}$. As X_{n_0} is closed, $x \in X_{n_0}$. Hence $x \in \bigcup_{n=1}^{N} X_n$. $\bigcup_{n=1}^{N} X_n$ is a closed subset of $\mathbb{R}^N$. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“闭包的序列判据与相对闭性”。完整定义、条件和论证见本讲原页第 76 页下方。

82 原课件第 82 页：闭包的序列判据与相对闭性

Limits in $\mathbb{R}^N$

Let S be a subset of $\mathbb{R}^N$; for a subset X of S, one defines the adherence of X in S as the set of points of S adherent to X. Similarly X is said to be closed in S if it owns the points of S adherent to X.

Example: (1) Let M be a positive real number, and take $S = \{x \in \mathbb{R}^2 \mid |x_n| < M \text{ for } n = 1, 2\}$, $X = \{x \in S \mid x_1 \geqslant 0\}$. From a sketch it will be clear that the set X is closed in S but not closed in $\mathbb{R}^2$. (2) Note also that any subset T of $\mathbb{R}^N$ is closed in T.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“闭包的序列判据与相对闭性”。完整定义、条件和论证见本讲原页第 76 页下方。

83 原课件第 83 页：闭包的序列判据与相对闭性



Substituting "the adherence in S" for "the adherence," "closed in S" for "closed," and S for $\mathbb{R}^N$ everywhere in (1) to (5), one obtains new theorems with the same proofs.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“闭包的序列判据与相对闭性”。完整定义、条件和论证见本讲原页第 76 页下方。

84 原课件第 84 页：任意欧氏子集都可分

Limits in $\mathbb{R}^N$

One can prove:

(6) Every subset S of $\mathbb{R}^N$ contains a countable subset X such that $S \subseteq \overline{X}$.

Proof: As $\overline{\mathbb{Q}^N} = \mathbb{R}^N$ and $\mathbb{Q}^N$ is countable, let $\mathbb{Q}^N = \{q^1, q^2, \cdots\}$ and

$$T = \left\{(n,k) \in \mathbb{N}^2 : S \cap B\left(q^n, \frac{1}{k}\right) \neq \varnothing\right\}.$$

For every $(n,k) \in T$, let

$$y_{n,k} \in S \cap B\left(q^n, \frac{1}{k}\right),$$

then

$$X = \{y_{n,k} : (n,k) \in T\}$$

is a countable subset of S. Then we need to show $S \subseteq \overline{X}$.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

枚举有理中心有理半径的球，对每个与 S 相交的球选择一个点 $s_j \in S$，形成可数集合 D。给定 $x \in S, \varepsilon > 0$，由可数基选球 B 满足 $x \in B \subset B(x,\varepsilon)$，其选点 s_j 与 x 距离小于 ε。故 $D \subset S \subset \overline{D}$。不能直接取 $S \cap \mathbb{Q}^N$，它可能为空，例如 $S = \{\sqrt{2}\}$。

85 原课件第 85 页：任意欧氏子集都可分

Limits in $\mathbb{R}^N$

Suppose any $x \in S$ and any $\varepsilon > 0$, there exists $k \in \mathbb{N}$ such that $\frac{2}{k} < \varepsilon$. Since $\overline{\mathbb{Q}^N} = \mathbb{R}^N$, there exists n so that

$$q^n \in B\left(x, \frac{1}{k}\right)$$

This implies $x \in B\left(q^n, \frac{1}{k}\right)$. So $(n, k) \in T$ and $y_{n,k} \in X$ and

$$d\left(x, y_{n,k}\right) < \frac{2}{k} < \varepsilon.$$

Hence $S \subseteq \overline{X}$. $\square$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“任意欧氏子集都可分”。完整定义、条件和论证见本讲原页第 84 页下方。

86 原课件第 86 页：任意欧氏子集都可分

Limits in $\mathbb{R}^N$

In other words, an arbitrary (therefore perhaps non-countable) subset S of $\mathbb{R}^N$ contains a countable set X which is dense in S, i.e., such that for any point x of S there are points of X arbitrarily close to x.

Example: Let S be $\mathbb{R}^N$ itself. From $\overline{\mathbb{Q}} = \mathbb{R}$ (example (2) of 1.6.h) follows $\overline{\mathbb{Q}^N} = \mathbb{R}^N$. Thus $\mathbb{Q}^N$, the product of N sets equal to $\mathbb{Q}$, i.e. the sets of points of $\mathbb{R}^N$ whose coordinates are all rationals, is dense in $\mathbb{R}^N$. Moreover, $\mathbb{Q}^N$ is countable. (apply (2) of 1.3.n to the N countable sets $\mathbb{Q}$).

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“任意欧氏子集都可分”。完整定义、条件和论证见本讲原页第 84 页下方。

87 原课件第 87 页：内部、边界与外部

Limits in $\mathbb{R}^N$

Let X be a subset of $\mathbb{R}^N$; a point $x \in \mathbb{R}^N$ is interior to X if it is not adherent to the complement of X. This means, intuitively, that x is completely surrounded by points of X. Any point x of $\mathbb{R}^N$ interior to X clearly belongs to X. The interior of X is the set of points interior to X, viz., $\mathbf{C}\left(\overline{\mathbf{C}X}\right)$.

Example: (1) Let M be a positive real numbers, and consider, in $\mathbb{R}^2$, the closed square $X = \left\{x \in \mathbb{R}^2 \mid |x_n| \leqslant M \text{ for } n = 1, 2\right\}$; its interior is $\left\{x \in \mathbb{R}^2 \mid |x_n| < M \text{ for } n = 1, 2\right\}$. (2) The set $\mathbb{Q}^2$ and the straight line $Y = \left\{y \in \mathbb{R}^2 \mid y_2 = 0\right\}$ in $\mathbb{R}^2$ have empty interiors.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

内部是 $\mathbb{R}^N \setminus \overline{E^c}$，外部是 $\mathbb{R}^N \setminus \overline{E}$，边界是 $\overline{E} \cap \overline{E^c}$。边界点的每个球都同时遇到集合及补集。三者互不交并覆盖空间；证明只需对一点分类：若有球全在 E 则内部，若有球全在补集则外部，否则边界。相对空间中所有补集和闭包都应相对 S 计算。

88 原课件第 88 页：内部、边界与外部

Limits in $\mathbb{R}^N$

As in the case of 1.6.I, if X is a subset of S (a subset of $\mathbb{R}^N$), a point $x \in S$ is said to be interior to X in S if it is not adherent to $\mathbf{C}_S X$, the complement of X in S.

Example: (3) Consider, in $\mathbb{R}^2$, the straight line $S = \{z \in \mathbb{R}^2 \mid z_1 = 0\}$ and its subset $Z = \{z \in \mathbb{R}^2 \mid z_1 = 0, 0 \leqslant z_2 \leqslant 1\}$. The interior of Z in S is $\{z \in \mathbb{R}^2 \mid z_1 = 0, 0 < z_2 < 1\}$; its interior in $\mathbb{R}^2$ is empty.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“内部、边界与外部”。完整定义、条件和论证见本讲原页第 87 页下方。

89 原课件第 89 页：内部、边界与外部

Limits in $\mathbb{R}^N$

Let X be a subset of $\mathbb{R}^N$; a point in $\mathbb{R}^N$ is a boundary point of X if it is adherent to both X and its complement $\mathbf{C}X$. The boundary of X is the set of its boundary points, viz., $\overline{X} \cap \overline{\mathbf{C}X}$.

Example: In the last group of examples, the boundary of the closed square X is its perimeter, that of $\mathbb{Q}^2$ is $\mathbb{R}^2$, that of Y is Y.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“内部、边界与外部”。完整定义、条件和论证见本讲原页第 87 页下方。

90 原课件第 90 页：内部、边界与外部



Limits in $\mathbb{R}^N$

Let X be a subset of $\mathbb{R}^N$; the exterior of X is the complement of its adherence, via., $\mathbf{C}\overline{X}$. It is easy to prove that the interior, the boundary, and the exterior of X form a partition of $\mathbb{R}^N$ or, what is equivalent, that the interior and the boundary of X form a partition of the adherence of X. The last remark shows that the set X is closed if and only if it contains its boundary.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“内部、边界与外部”。完整定义、条件和论证见本讲原页第 87 页下方。

91 原课件第 91 页：有界、紧与有限积

Limits in $\mathbb{R}^N$

A subset S of $\mathbb{R}^N$ is said to be bounded if it is contained in some closed cube $\mathbb{M}$. A set which is unbounded is therefore a set which has points arbitrarily far from the origin.

Example: The set $\mathbb{J}$ of integers (a subset of $\mathbb{R}$) is unbunded.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

有界是存在统一 M 使每个点的范数不大于 M。有限积的闭性由坐标收敛和各因子闭性给出；有界性由各因子界的平方和给出。因此有限个紧集的积紧。有限维条件很关键，无限维中闭且有界一般不保证紧。

92 原课件第 92 页：有界、紧与有限积

Limits in $\mathbb{R}^N$

A subset S of $\mathbb{R}^N$ is said to be compact if it is closed and bounded. It is easily proved that:

(7) If $S_1, \cdots, S_k, \cdots, S_K$ are closed (resp. compact) subsets of $\mathbb{R}^{N_1}, \cdots, \mathbb{R}^{N_k}, \cdots, \mathbb{R}^{N_K}$, then $S_1 \times \cdots \times S_k \times \cdots \times S_K$ is a closed (resp. compact) subset of $\mathbb{R}^{\sum_{k=1}^{K} N_k}$.

Proof: Let $x = (x_1, \cdots, x_k, \cdots, x_K) \in S_1 \times \cdots \times S_k \times \cdots \times S_K$ where $x_k \in S_k$ for $k = 1, \cdots, K$.

For any $x \in \overline{S_1 \times \cdots \times S_k \times \cdots \times S_K}$, there exists $\{x^q; q \in \mathbb{N}\} \subseteq S_1 \times \cdots \times S_k \times \cdots \times S_K$ such that $x^q \to x$. Hence for any $k = 1, \cdots, K$, we have $x_k^q \to x_k$. As S_k is closed, $x_k \in S_k$. Hence $x \in S_1 \times \cdots \times S_k \times \cdots \times S_K$. So $S_1 \times \cdots \times S_k \times \cdots \times S_K$ is a closed.

If $S_1, \cdots, S_k, \cdots, S_K$ are compact, then for $k = 1, \cdots, K$, any $x_k \in S_k$, there exists $M_k \geqslant 0$ such that $\|x_k\| \leqslant M_k$. Let $M = \max\{M_1, \cdots, M_k, \cdots, M_K\}$, then for any $x = (x_1, \cdots, x_k, \cdots, x_K) \in S_1 \times \cdots \times S_k \times \cdots \times S_K$, we have $\|x\| \leqslant M$, i.e. $S_1 \times \cdots \times S_k \times \cdots \times S_K$ is bounded. As $S_1 \times \cdots \times S_k \times \cdots \times S_K$ is also closed, we have that $S_1 \times \cdots \times S_k \times \cdots \times S_K$ is compact. □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“有界、紧与有限积”。完整定义、条件和论证见本讲原页第 91 页下方。

93 原课件第 93 页：连通的多种判据



Limits in $\mathbb{R}^N$

Finally, a subset S of $\mathbb{R}^N$ is said to be connected if it is not the union of two non-empty, disjoint subsets closed in S. In other words, a set S is connected if it cannot be partitioned into two non-empty subsets closed in S, i.e., loosely speaking, if it is of one piece (but perhaps with holes). One can prove that:

(8) A subset of $\mathbb{R}$ is connected if and only if it is an interval.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

连通指不能分成两个非空不交的相对开集；取补立刻等价于不能分成两个非空不交的相对闭集，也等价于没有非平凡既开又闭子集。$\partial A = \overline{A} \setminus A^\circ = \varnothing$ 等价于 A 既开又闭，故得边界判据。若不交集合 A, B 覆盖空间且互不碰对方闭包，它们在空间中均闭也均开，给分离判据。最后，非平凡 clopen 集 A 的示性函数 $1_A : X \to \{0,1\}$ 连续，因为每个离散开集的逆像均开；反之非常值连续函数到 $\{0,1\}$ 的两个纤维给分离。实数连通集必是区间：若中间缺点 c，两侧相对开集分离它。区间必连通：若分离且 a 在一侧、$b > a$ 在另一侧，取 $s = \sup\{t \in [a,b] : t \text{ 在第一侧}\}$，利用任一侧的相对开性都矛盾。

94 原课件第 94 页：连通的多种判据

Limits in $\mathbb{R}^N$

Proof: Let A is connected. Assume A is not an interval, then we can take three real numbers a_1, a_2, a such that $a_1 < a < a_2$, $a_1 \in A$, $a_2 \in A$ and $a \notin A$. Let $A_1 = A \cap (-\infty, a]$, $A_2 = A \cap [a, \infty)$, then $A = A_1 \cup A_2$ and $A_1 \cap A_2 = \varnothing$. Notice that $(-\infty, a]$ and $[a, \infty)$ are both non-empty closed subsets of $\mathbb{R}$, hence A_1 and A_2 are closed in A, hence A is therefore not connected. Contradiction!

Conversely let A be an interval of $\mathbb{R}$. To derive a contradiction, suppose we could write $A = A_1 \cup A_2$ for nonempty and disjoint closed subsets A_1 and A_2 of A. Pick any $a_1 \in A_1$ and $a_2 \in A_2$ and let $a_1 < a_2$ without loss of generality. Define $a = \sup\{t \in A_1 : t < a_2\}$ [or $a = \inf\{t \in A_2 : a_1 < t\}$], and note that $a_1 \leqslant a \leqslant a_2$ and hence $a \in A$ (because A is an interval).

If $a \in A_1$, then $a \neq a_2$ (since $A_1 \cap A_2 = \varnothing$), so $a < a_2$. As $a \notin A_2$ and A_2 is closed, then we know there exists $\varepsilon > 0$ such that for all $x \in A_2$, we have $|x - a| > \varepsilon$. Hence for all $x \in (a, a + \varepsilon)$, $a_1 < x < a_2$, hence $x \notin A_1 \cup A_2$, i.e. $x \notin A$. Contradiction!

If $a \in A_2$, then $a \neq a_1$ (since $A_1 \cap A_2 = \varnothing$), so $a_1 < a$. As $a \notin A_1$ and A_1 is closed, then we know there exists $\varepsilon > 0$ such that for all $x \in A_1$, we have $|x - a| > \varepsilon$. Hence for all $x \in (a - \varepsilon, a)$, $a_1 < x < a_2$, hence $x \notin A_1 \cup A_2$, i.e. $x \notin A$. Contradiction! $\square$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“连通的多种判据”。完整定义、条件和论证见本讲原页第 93 页下方。

95 原课件第 95 页：连通的多种判据

Connectedness

The two sets A_1 and A_2 in metric space X are said to be separated if

$$(A_1 \cap \overline{A}_2) \cup (A_2 \cap \overline{A}_1) = \varnothing$$

If there exist two non-empty separated sets A_1 and A_2 in metric space X such that $X = A_1 \cup A_2$, then X is a disconnected space. Otherwise, X is a connected space, i.e., there are not two non-empty separated sets A_1 and A_2 in metric space X such that $X = A_1 \cup A_2$.

Specifically, a subset S of metric space X is said to be connected if it is not the union of two non-empty separated sets in S. In other words, a set S is connected if it cannot be partitioned into two non-empty separated sets in S, i.e., loosely speaking, if it is of one piece (but perhaps with holes).

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“连通的多种判据”。完整定义、条件和论证见本讲原页第 93 页下方。

96 原课件第 96 页：连通的多种判据

Connectedness

(8) A subset A of $\mathbb{R}$ is connected if and only if it is an interval.

Proof. Let A is connected. Assume A is not an interval, then we can take three real numbers a_1, a_2, a such that $a_1 < a < a_2$, $a_1 \in A$, $a_2 \in A$ and $a \notin A$. Let $A_1 = A \cap (-\infty, a]$, $A_2 = A \cap [a, \infty)$, then $A = A_1 \cup A_2$ and $A_1 \cap A_2 = \varnothing$. Notice that $(-\infty, a]$ and $[a, \infty)$ are both non-empty closed subsets of $\mathbb{R}$, hence A_1 and A_2 are closed in A, hence A is therefore not connected. Contradiction!
Conversely let A be an interval of $\mathbb{R}$. To derive a contradiction, suppose we could write $A = A_1 \cup A_2$ for nonempty and disjoint closed subsets A_1 and A_2 of A. Pick any $a_1 \in A_1$ and $a_2 \in A_2$ and let $a_1 < a_2$ without loss of generality. Define $a = \sup\{t \in A_1 : t < a_2\}$, and note that $a_1 \leqslant a \leqslant a_2$ and hence $a \in A$ (because A is an interval).
If $a \in A_1$, then $a \neq a_2$ (since $A_1 \cap A_2 = \varnothing$), so $a < a_2$. As $a \notin A_2$ and A_2 is closed, then we know there exists $\varepsilon > 0$ such that for all $x \in A_2$, we have $|x - a| > \varepsilon$. Hence for all $x \in (a, a + \varepsilon)$, $a_1 < a < a_2$, hence $x \notin A_1 \cap A_2$, i.e. $x \notin A$. Contradiction!
If $a \in A_2$, then $a > a_1$. Using similar method, we can also get to the contradiction! □

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“连通的多种判据”。完整定义、条件和论证见本讲原页第 93 页下方。

97 原课件第 97 页：连通的多种判据

Connectedness

Theorem: For a metric space X, the following statements are equivalent:

(1) X cannot be divided into two disjoint non-empty open sets.

(2) X cannot be divided into two disjoint non-empty closed sets. [That is, X is connected.]

(3) The only clopen subsets of X are X and the empty set. (A clopen set is both open and closed)

(4) The only subsets of X with empty boundary are X and the empty set.

(5) X cannot be written as the union of two non-empty separated sets (sets for which each is disjoint from the other's closure).

(6) All continuous functions from X to $\{0,1\}$ are constant, where $\{0,1\}$ is the two-point space endowed with the discrete metric.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“连通的多种判据”。完整定义、条件和论证见本讲原页第 93 页下方。

98 原课件第 98 页：连通的多种判据

Connectedness

Proof of (1) $\Longrightarrow$ (2): If (2) is false, then there exists non-empty, disjoint and closed sets A_1 and A_2 such that $X = A_1 \cup A_2$. Then $\varnothing = A_1 \cap A_2$, hence $X = \mathbf{C}\varnothing = \mathbf{C}A_1 \cup \mathbf{C}A_2$ where $\mathbf{C}A_1$ and $\mathbf{C}A_2$ are disjoint, non-empty and open sets. Thus (1) is false.
Proof of (2) $\Longrightarrow$ (1): If (1) is false, then there exists non-empty, disjoint and open sets A_1 and A_2 such that $X = A_1 \cup A_2$. Then $\varnothing = A_1 \cap A_2$, hence $X = \mathbf{C}\varnothing = \mathbf{C}A_1 \cup \mathbf{C}A_2$ where $\mathbf{C}A_1$ and $\mathbf{C}A_2$ are disjoint, non-empty and closed sets. Thus (2) is false.
Proof of (1) $\Longrightarrow$ (3) and (2) $\Longrightarrow$ (3): If the clopen subset S of X is a proper subset ($S \notin \{\varnothing, X\}$), then $X = S \cup \mathbf{C}S$, where $\mathbf{C}S$ is a clopen proper subset of X, thus X is not connected.
Proof of (3) $\Longrightarrow$ (1): If (1) is false, then there exist non-empty, disjoint and open sets A_1 and A_2 such that $X = A_1 \cup A_2$, hence $A_2 = \mathbf{C}A_1$ is closed. Thus A_2 a clopen proper subset of X, that is, (3) is false.
Proof of (3) $\Longrightarrow$ (2): If (2) is false, then there exist non-empty, disjoint and closed sets A_1 and A_2 such that $X = A_1 \cup A_2$, hence $A_2 = \mathbf{C}A_1$ is open. Thus A_2 a clopen proper subset of X, that is, (3) is false.

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“连通的多种判据”。完整定义、条件和论证见本讲原页第 93 页下方。

99 原课件第 99 页：连通的多种判据

Connectedness

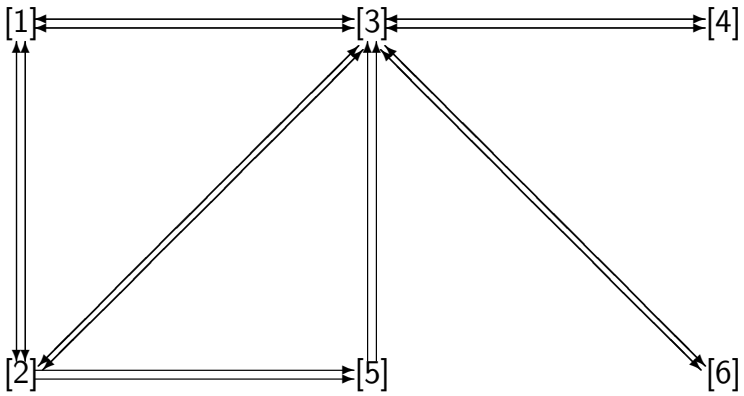
Proof of (3) $\Longleftrightarrow$ (4): $A \subseteq X$ with $\partial A = \varnothing$ if and only if A is a clopen subset of X.
$A \subseteq X$ with $\partial A = \varnothing$ if and only if $\overline{A} = \overset{\circ}{A} \cup \partial(A)$ with $\partial A = \varnothing$ if and only if $\overline{A} = \overset{\circ}{A}$ if and only if $\overline{A} = A = \overset{\circ}{A}$ if and only if A is a clopen subset of X.
Proof of (2) $\Longrightarrow$ (5): If (5) is false, then there exist non-empty separated sets A_1 and A_2 such that $X = A_1 \cup A_2$. Since $\overline{A_1} \cap A_2 = \varnothing$ and $\overline{A_2} \cap A_1 = \varnothing$, then $\overline{A_1} \subseteq A_1$ and $\overline{A_2} \subseteq A_2$, i.e. A_1 and A_2 are non-empty closed. Thus (2) is false.
Proof of (5) $\Longrightarrow$ (3): If (3) is false, then there exist a non-empty clopen proper subset A of X, hence $X = A \cup \mathbf{C}A$ where $\mathbf{C}A$ is a non-empty clopen proper subset of X. Thus A and $\mathbf{C}A$ are non-empty separated sets, that is, (5) is false.
Proof of (3) $\Longrightarrow$ (6): As $\{0\}$ and $\{1\}$ are clopen proper subsets in the space $\{0, 1\}$, two-point space endowed with the discrete metric. Then if f is continuous, $f^{-1}(\{0\})$ and $f^{-1}(\{1\})$ are clopen subsets, then $f^{-1}(\{0\})$ and $f^{-1}(\{1\})$ are either X or $\varnothing$, which implies f is constant.
Proof of (6) $\Longrightarrow$ (3): As $\{0\}$ and $\{1\}$ are clopen proper subsets in the space $\{0, 1\}$, two-point space endowed with the discrete metric. Then if f is continuous, $f^{-1}(\{0\})$ and $f^{-1}(\{1\})$ are clopen subsets. Then if f is constant, $f^{-1}(\{0\})$ and $f^{-1}(\{1\})$ are either X or $\varnothing$. Therefore the only clopen subsets of X are X and the empty set. $\square$

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“连通的多种判据”。完整定义、条件和论证见本讲原页第 93 页下方。

100　原课件第 100 页：连通的多种判据

Connectedness



上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页继续“连通的多种判据”。完整定义、条件和论证见本讲原页第 93 页下方。